

Project Code	AI-Agents
Type	DATA601 (Academic research project)
Title	Automated Customer Support Ticket Analysis and Response System
Department	University of Canterbury (Maths and Stats)
Location	University of Canterbury
Students	Group

Project Summary

Project Overview

Leverage a Large Language Model (LLM) and multiple AI agents to analyze customer support tickets, categorize them, prioritize issues, and generate tailored responses. This system can help businesses streamline their customer support process by automating ticket triage and response generation while incorporating sentiment analysis and escalation protocols.

Why this Project?

- **Practical Application:** Automating customer support is a real-world use case with measurable business impact.
- **LLM Strengths:** Leverages the LLM's natural language understanding for classification, sentiment analysis, and response generation.
- **Agent Collaboration:** Demonstrates multi-agent systems working together (classification, sentiment, response, escalation).
- **Scalability:** Can be extended to include real-time web searches (e.g., using Grok 3's DeepSearch mode) for product-specific solutions or integration with CRMs like Salesforce or AwareIM.
- **Learning Outcomes:** Combines NLP, agent-based systems, data pre-processing, modelling and evaluation metrics, making it a comprehensive end-to-end data science project.

LLM and Agent Utilization

- **LLM Role:** The LLM (e.g., ChatGPT, Grok 3) processes incoming customer support tickets (text data) to understand the context, extract key information (e.g., issue type, urgency, customer sentiment), and generate human-like responses.
- **Agent Roles:**
 - **Classifier Agent:** Analyzes the ticket text to categorize it (e.g., technical issue, billing query, feature request) and assigns a priority level based on keywords and sentiment.
 - **Sentiment Analysis Agent:** Uses the LLM to assess the emotional tone of the ticket (e.g., angry, neutral, satisfied) to flag urgent or sensitive cases for human escalation.
 - **Response Generator Agent:** Crafts personalized, context-aware responses using the LLM, incorporating company tone and policies.
 - **Escalation Manager Agent:** Monitors categorized tickets and sentiment scores to route high-priority or complex issues to human agents, including a summary of the ticket for quick handover.

Implementation Steps

The following is a suggested approach to implementation.

- **Data Collection:** Gather a dataset of customer support tickets (e.g., from open-source datasets like Zendesk or Kaggle, or simulate synthetic data using an LLM). Here are some URL's to use as a starting point:
 - <https://www.kaggle.com/datasets/suraj520/customer-support-ticket-dataset>
 - <https://www.kaggle.com/datasets/tobiasbueck/multilingual-customer-support-tickets>

- <https://github.com/gabrielpreda/Support-Tickets-Classification>
- **Pre-processing:** Clean the ticket text (remove noise, standardize formats, tokenization, stemming/lemmatization, stopword removal) and label a subset for training/validation/evaluation (e.g., categories, sentiment).
- **Agent Workflow:**
 - **Classifier Agent:** Fine-tune the LLM or use prompt engineering to classify tickets into predefined categories (e.g., "bug," "billing," "general inquiry") and assign urgency scores.
 - **Sentiment Agent:** Use LLM prompts or fine-tune with Hugging Face Transformers for sentiment analysis, scoring tickets on a scale (e.g., negative, neutral, positive).
 - **Response Agent:** Prompt the LLM to generate responses based on ticket category, sentiment, and company guidelines (e.g., "We apologize for the inconvenience. Our team is investigating your issue...").
 - **Escalation Agent:** Define rules for human escalation based on urgency/sentiment and generate concise summaries using the LLM for quick human review.
- **Integration:** Create a pipeline where agents interact (e.g., Classifier passes data to Sentiment, which informs Response and Escalation). Use a framework like LangChain or AutoGen to manage agent coordination. A visual flowchart is recommended.
- **Evaluation:** Measure performance using metrics like classification accuracy (for categorization), sentiment prediction accuracy, and response quality (human evaluation or BLEU/ROUGE scores). Set clear benchmarks or performance thresholds for acceptability. Hint: Are your classes balanced? if not, is accuracy a valid metric?
- **Deployment:** Build a simple interface (e.g., Flask, Shiny or Streamlit) where tickets are input, processed by agents, and responses/escalations are displayed. Consider basic usability principles (clear visualization, ease of interaction).

Team Roles and Collaboration

Here's a proposed role breakdown, which may vary based on the team's skills and strengths.

- **Student 1 (Data Engineer):** Oversees data collection (e.g., sourcing datasets from Kaggle or generating synthetic data) and preprocessing (cleaning, tokenization, standardization). Ensures data quality and prepares labeled subsets for training and evaluation.
- **Student 2 (NLP Specialist):** Designs and implements LLM prompts or fine-tuning for the Classifier, Sentiment Analysis, and Response Generator agents. Validates the semantic accuracy of ticket categorization, sentiment scoring, and response quality.
- **Student 3 (Agent Systems Developer):** Builds and coordinates the multi-agent workflow using frameworks like LangChain or AutoGen. Ensures seamless interaction between the Classifier, Sentiment, Response, and Escalation agents, including rule-based logic for escalation.
- **Student 4 (UI Developer):** Develops a user-friendly interface (e.g., using Flask, Shiny, or Streamlit) to input tickets, display agent outputs (categories, sentiment, responses), and show escalation summaries. Focuses on clear visualization and usability.

Collaboration: All team members work together on integrating the pipeline (e.g., connecting preprocessed data to agents, linking agent outputs to the UI) and evaluating the system's performance (e.g., classification accuracy, response quality).

Timeline

Here's a proposed project timeline.

- **Weeks 1-2:** Project planning, dataset selection (e.g., Kaggle or synthetic data), and environment setup (LLM access, agent frameworks, Python libraries).
- **Weeks 3-4:** Data collection and preprocessing (cleaning, tokenization, labeling for categories and sentiment).

- Weeks 5-6: Development of individual agents (Classifier, Sentiment Analysis, Response Generator, Escalation Manager) using LLM prompts or fine-tuning.
- Weeks 7-8: Implementation of agent workflow coordination (e.g., using LangChain or AutoGen) and initial UI development (e.g., Flask, Shiny, or Streamlit).
- Weeks 9-10: System integration (connecting data pipeline, agents, and UI), testing (e.g., classification accuracy, sentiment scoring, response quality), and debugging.
- Weeks 11-12: Final refinements, performance evaluation (e.g., BLEU/ROUGE scores, human review), documentation, and presentation preparation.

Tools and Technologies

These are suggested tools and technologies; feel free to use alternatives that you find more suitable or are more comfortable with.

- LLM: Grok 3 (accessible via grok.com or xAI API for higher quotas; see <https://x.ai/api> for details).
- Agent Framework: LangChain, AutoGen, or custom Python scripts for agent coordination.
- Data Processing: Pandas, NumPy for ticket pre-processing.
- Sentiment Analysis: Use LLM prompts or fine-tune with libraries like Hugging Face's Transformers.
- Visualization: A simple user interface e.g. Flask, Shiny or Streamlit, to visualize interesting aspects of your data.

Learning Outcomes

Technical Skills:

- Master data pre-processing techniques for text data, including cleaning, tokenization, and labeling for NLP tasks.
- Develop proficiency in LLM prompt engineering and fine-tuning for tasks like ticket classification, sentiment analysis, and response generation.
- Gain experience in designing and coordinating multi-agent systems using frameworks like LangChain or AutoGen.
- Learn to implement and evaluate NLP models using metrics such as classification accuracy, F1-score, and BLEU/ROUGE for response quality.
- Build practical skills in creating user-friendly interfaces with tools like Streamlit, Flask, or Shiny for data-driven applications.

Soft Skills:

- Enhance teamwork and collaboration through role-based contributions and integrated system development.
- Strengthen project management abilities by adhering to milestones, managing timelines, and coordinating tasks.
- Improve communication skills through clear documentation, presentations, and discussions of technical and ethical considerations.

Ethical and Professional Skills:

- Understand ethical implications of automated customer support, including data privacy, bias mitigation, and transparency in AI decision-making.
- Develop critical thinking in evaluating system performance and addressing real-world challenges like class imbalance or response appropriateness.

Portfolio Impact:

- Create a showcase project demonstrating end-to-end data science skills, from data pre-processing to agent-based automation and UI deployment, with strong industry relevance.

Resources

Here are a list of resources that you may find as useful starting points for your research.

Open-Source Datasets:

- Kaggle: Customer Support Ticket Dataset (<https://www.kaggle.com/datasets/suraj520/customer-support-ticket-dataset>).
- Kaggle: Multilingual Customer Support Tickets (<https://www.kaggle.com/datasets/tobiasbueck/multilingual-customer-support-tickets>).
- GitHub: Support Tickets Classification (<https://github.com/gabrielpreda/Support-Tickets-Classification>).

LLM and Agent Frameworks:

- LangChain Documentation: Guides for building multi-agent systems (<https://python.langchain.com/docs/>).
- AutoGen Documentation: Tutorials for agent coordination (<https://microsoft.github.io/autogen/>).
- xAI API: Access to Grok 3 for LLM tasks (<https://x.ai/api>).

NLP Tools:

- Hugging Face Transformers: Pre-trained models and fine-tuning for sentiment analysis (<https://huggingface.co/docs/transformers/>).
- NLTK/Spacy: Text preprocessing (tokenization, lemmatization) (<https://www.nltk.org/>, <https://spacy.io/>).

UI Development:

- Streamlit Documentation: Building interactive web apps (<https://docs.streamlit.io/>).
- Flask Tutorials: Lightweight web app development (<https://flask.palletsprojects.com/>).
- Shiny for Python: Interactive UI for data apps (<https://shiny.posit.co/py/>).

Evaluation Metrics:

- Scikit-learn: Classification metrics like accuracy and F1-score (<https://scikit-learn.org/stable/>).
- NLTK: BLEU/ROUGE scores for response quality (<https://www.nltk.org/>).

Deliverables

For detailed requirements of the project report and presentation, please consult LEARN.

- A working automated customer support ticket analysis and response system, hosted locally or on a cloud platform (e.g., Streamlit Sharing, Render for Flask apps), capable of processing tickets, categorizing them, analyzing sentiment, generating responses, and flagging escalations.
- A final report detailing the methodology (data preprocessing, agent design, workflow integration), evaluation metrics (e.g., classification accuracy, sentiment prediction, response quality), challenges, and results.
- A presentation showcasing the system, demonstrating ticket processing (e.g., input, categorization, response generation, escalation), and discussing technical insights and learning outcomes.

Optional: A visualization of ticket data (e.g., sentiment distribution, category breakdown) to highlight patterns or system performance.

Potential Challenges and Mitigations

- Data Quality and Availability: Incomplete or noisy customer support ticket datasets may affect system performance.
 - Mitigation: Thoroughly clean datasets during pre-processing (e.g., remove duplicates, handle missing values) and supplement with synthetic data generated by an LLM if needed.

- **Agent Coordination Complexity:** Ensuring seamless interaction between Classifier, Sentiment, Response, and Escalation agents may be challenging.
 - Mitigation: Use a framework like LangChain or AutoGen to streamline agent workflows and test interactions incrementally to identify bottlenecks.
- **LLM Performance Variability:** Inconsistent LLM outputs (e.g., misclassified tickets, inappropriate responses) may reduce system reliability.
 - Mitigation: Refine prompts through iterative testing, fine-tune the LLM on labeled data if possible, and establish clear performance benchmarks (e.g., accuracy, BLEU scores).
- **Class Imbalance in Ticket Categories:** Uneven distribution of ticket types (e.g., more billing queries than feature requests) may skew classification accuracy.
 - Mitigation: Use evaluation metrics like F1-score instead of accuracy and apply techniques like oversampling or weighted loss functions to balance classes.
- ****UI Usability and Integration**:** Developing a user-friendly interface that effectively displays agent outputs and escalations may be time-intensive.
 - Mitigation: Leverage pre-built UI frameworks like Streamlit or Flask, prioritize simple and clear visualizations, and conduct user testing to ensure usability.
- **Ethical Concerns:** Automated responses or escalations may inadvertently introduce bias or fail to address sensitive customer issues appropriately.
 - Mitigation: Implement bias detection during sentiment analysis, ensure transparency in automated decisions, and include human oversight for escalated cases.
- **Team Coordination and Time Management:** Balancing workloads and meeting deadlines across diverse tasks (data, agents, UI) may be difficult.
 - Mitigation: Use tools like GitHub for code collaboration and Trello/Slack for task tracking, and break tasks into clear milestones with assigned responsibilities.

Scalability and Extensions

Here are some ideas for stretch goals if time permits.

1. Add a feedback loop where customer responses to generated replies are analyzed to improve the system.
2. Integrate with X posts to monitor real-time customer sentiment (requires X API access).
3. Use Grok 3's voice mode (available on iOS/Android apps) to convert tickets to voice summaries for human agents.

Ethical considerations

- Address data privacy, bias detection, transparency in automated decision-making.
- Discuss the importance of responsible AI and clear communication with users.
- **Documentation and Iterative Improvements**
- Students should document challenges, solutions, and iterative refinements to demonstrate the development process clearly, enhancing learning outcomes and project transparency.

Summary

This project balances technical complexity with practical utility, showcasing the power of LLMs and agent-based workflows in data science.

Project Code	Bodeker
Type	DATA605 (Industry based research project)
Title	Data-Driven Prediction of Sunlight and Indoor Temperature to Improve Smart Façade Operation
Industry partner	Bodeker
Location	University
Students	Individual

Project Summary

Overview

Bodeker are looking for a motivated student data scientist or engineer to participate in a team that is developing simple, data-driven models to forecast short-term sunlight and indoor temperature. These forecasts will be used to automate façade operation (e.g., opening or closing blinds) to improve indoor comfort and reduce energy use.

Key Responsibilities

- Collect and clean building data and weather data.
- Build and test short-term models to forecast key weather parameters related to building thermal load.
- Design simple control rules for façade operation.
- Evaluate comfort and performance improvements.

Skills and Tools

- Python (pandas, numpy, scikit-learn).
- Basic machine learning and time-series analysis.
- Data visualisation.
- Interest in sustainable building technologies.

Data Sources

- Open building and weather datasets (e.g., solar radiation, indoor temperature).

Timeline

- 10-week project including data prep, model building, control testing, and reporting.

Deliverables

- Working Python notebook with models and control logic.
- Graphs showing prediction results and comfort impact.
- Final report with key findings (see LEARN).

Project Code	Canopy
Type	DATA601 (Academic research project)
Title	Tree Canopy Cover of the University of Canterbury Campus
Department	Sustainability Office Facilities Management
Location	University of Canterbury
Students	Group

Project Summary

Background

The University of Canterbury's Biodiversity Plan outlines identified targets and actions that support the University's Sustainability Strategy. Annual University of Canterbury Sustainability Reports contain reporting against this Biodiversity Plan. This is used to evaluate the university's quality and rank it under various international systems.

The underlying principle of the UC Biodiversity Plan's approach is that the more diverse the biological life on campus, the more ecologically healthy it will be. This not only helps the University meet its environmental obligations but also enables the delivery of core research and teaching opportunities.

The UC Biodiversity Plan sets a target to measure and increase tree canopy cover. From 2015/2016 and 2018/2019 data, Associate Professor Justin Morgenroth investigated and reported on the tree canopy cover of the city of Christchurch, narrowing that down to provide the baseline for canopy cover on the Ilam campus, [linked here to the full report](#).

In the summer of 2024, an Applied Data Science student completed a tree canopy cover estimate for the University of Canterbury using three data sets of aerial photographs and LiDAR from 2015/2016, 2018/2019, and 2020/2021 as input features in object-based image analysis. This work concludes by recommending the exploration of improving classification accuracy.

Project Brief

We expect an in-depth search for alternative, more recent data sets and subsequently run those, if available, to build the changing picture over time. If new data is available, report on the change to canopy cover over the time lapse. This will involve replicating the same work carried out by the previous student. The canopy cover was estimated using an object-based image analysis (OBIA) approach, supplemented with manual corrections. The OBIA was based on aerial photography and LiDAR data acquired over Christchurch during the summer of 2015/2016, 2018/2019 and 2020/2021. Most importantly, to explore the recommendations to improve the classification accuracy of the deep learning model.

Alongside this tree canopy cover metric, we are also establishing baselines for various aspects to gauge and measure biodiversity on campus. This includes delineating the land use in gardens, lawns, water bodies, and the built environment. Therefore, it would also be requested to scope and, if feasible, carry out mapping of permeable surfaces, delineating land use on campus, and mapping areas of lawns, gardens, buildings, and other built environments.

Project Code	CiJR
Type	DATA601 (Academic research project)
Title	Data-Driven Modelling of Head Impacts for Brain Strain Estimation
Department	University of Canterbury, School of Psychology, Speech and Hearing
Location	University of Canterbury
Students	Individual

Project Summary

Background

This project builds on a growing body of research aimed at improving our understanding of how head impacts during sport translate into brain strain - a key factor in assessing concussion and traumatic brain injury risk. The work leverages real-world data collected from instrumented mouthguards worn by female rugby players, capturing detailed linear and angular acceleration measurements during impacts. Led by Professor Nick Draper in the CiJR research group at the University of Canterbury, the project draws from his extensive background in sport and exercise science, with a strong emphasis on translational research - turning data into practical solutions. The current research focus is on developing surrogate machine learning models that can predict brain strain based on measurable impact kinematics, eliminating the need for computationally expensive finite element simulations. This innovation has the potential to enable rapid, scalable assessments of brain injury risk in athletic contexts.

Problem statement

This project aims to advance the understanding of how head impacts translate into brain strain, a critical factor in concussion and traumatic brain injury research. The overarching goal is to develop surrogate models that use measurable impact kinematics (e.g. peak linear and angular accelerations, full-time series signals) to predict brain strain values without the implementation of a finite element model. This will allow for rapid, scalable brain injury risk assessment without requiring computationally expensive simulations.

Expected methodology

A MADS student joining this project will have the opportunity to apply and demonstrate their data science and machine learning expertise through a range of activities. These include organizing and cleaning raw head impact datasets collected across multiple athletes, developing pipelines for automated feature extraction from impact kinematics, and training and validating machine learning models, such as random forests, neural networks, or gradient boosting, to predict brain strain metrics. The student will also create interactive visualizations or dashboards to compare impact profiles across athletes.

Deliverables

By the end of the project, the student should aim to build a reproducible data pipeline for managing complex multi-athlete, multi-impact datasets, implement baseline predictive models for brain strain estimation (using metrics such as UBrIC or other derived features), and benchmark model performance using cross-validation and appropriate error metrics.

Data / resources provided

We are working with datasets collected from deployed mouthguards in youth rugby players, these datasets include linear and angular acceleration measurements from athletes during real-world impacts.

Project Code	CustSeg
Type	DATA601 (Academic research project)
Title	Customer Segmentation
Department	University of Canterbury (Maths and Stats)
Location	University of Canterbury
Students	Group

Project Summary

Project Overview

This project aims to develop a customer segmentation system using clustering techniques and a Large Language Model (LLM) to group customers based on their purchasing behavior, demographics, and preferences. By analyzing datasets from Kaggle (or similar), the system will identify distinct customer segments to inform targeted marketing strategies, improve customer retention, and optimize business decisions. A multi-agent system will preprocess data, perform clustering, interpret segments, and generate actionable marketing recommendations.

Why this Project?

- **Practical Application:** Customer segmentation is widely used in industries like retail and e-commerce to personalize marketing and enhance customer experiences.
- **LLM Strengths:** Leverages LLMs for interpreting complex customer data and generating human-readable segment descriptions and marketing strategies.
- **Agent Collaboration:** Demonstrates multi-agent systems for data pre-processing, clustering, and recommendation generation.
- **Scalability:** Can integrate real-time data (e.g., via Grok 3’s DeepSearch mode) or connect to CRM systems for deployment.

Learning Outcomes: Combines unsupervised learning, NLP, data preprocessing, and visualization, offering a comprehensive data science experience.

LLM Role: The LLM (e.g., Grok 3) processes textual customer data (e.g., reviews, feedback) to extract features like sentiment or preferences and generates natural language descriptions of customer segments and marketing recommendations.

Agent Roles:

- **Preprocessing Agent:** Cleans and transforms customer data (e.g., handling missing values, encoding categorical variables, normalizing numerical features).
- **Clustering Agent:** Applies clustering algorithms (e.g., K-means, DBSCAN) to group customers based on features like purchase history, age, or spending patterns.
- **Interpretation Agent:** Uses the LLM to analyze clusters and generate descriptive profiles (e.g., “Young, high-spending tech enthusiasts”).
- **Recommendation Agent:** Crafts tailored marketing strategies for each segment using the LLM, incorporating business objectives and tone.

Implementation Steps

The following is a suggested approach to implementation:

1. **Data Collection:** Obtain customer datasets from Kaggle, such as:
 - <https://www.kaggle.com/datasets/vjchoudhary7/customer-segmentation-tutorial-in-python>

- <https://www.kaggle.com/datasets/kandij/mall-customers>
 - <https://www.kaggle.com/datasets/imakash3011/customer-personality-analysis>
 - <https://www.kaggle.com/code/halimedogan/customer-segmentation-using-rfm/input>
2. **Pre-processing:** Clean data (handle missing values, outliers), encode categorical variables, normalize numerical features, and extract text-based features (e.g., sentiment from reviews) using an LLM or more traditional manual techniques.
 3. **Clustering:**
 - Apply clustering algorithms (e.g., K-means, hierarchical clustering) to segment customers.
 - Use dimensionality reduction (e.g., ICA, PCA, t-SNE) for visualization and feature optimization.
 4. **Interpretation:** Prompt the LLM to generate human-readable descriptions of each cluster based on feature distributions and customer profiles.
 5. **Recommendation:** Use the LLM to create marketing strategies for each segment (e.g., discounts for price-sensitive customers, premium offers for high-spenders).
 6. **Evaluation:** Assess clustering quality using metrics like silhouette score, Davies-Bouldin index, and business relevance (e.g., segment distinctiveness). Validate LLM-generated outputs via human review.
 7. **Visualization and Deployment:** Build an interactive interface (e.g., Shiny, Streamlit, Flask) to display segment profiles, visualizations (e.g., cluster scatter plots), and marketing recommendations.

Team Roles and Collaboration

Here's a proposed role breakdown, which may vary based on the team's skills and strengths.

- **Data Engineer:** Leads data collection, preprocessing, and feature engineering to ensure high-quality input for clustering.
- **Machine Learning Specialist:** Implements and optimizes clustering algorithms, evaluates cluster quality, and applies dimensionality reduction.
- **NLP Specialist:** Designs LLM prompts for feature extraction (e.g., sentiment), cluster interpretation, and recommendation generation.
- **UI Developer:** Creates an interactive interface to visualize segments and display LLM-generated recommendations.
- **Collaboration:** All members collaborate on integrating components (e.g., connecting clustering outputs to LLM interpretation, linking results to UI) and testing the end-to-end system. Use tools like GitHub and Trello for coordination.

Timeline

Here's a proposed project timeline:

- **Weeks 1-2:** Project planning, dataset selection, and environment setup (e.g., Python, LLM access, clustering libraries).
- **Weeks 3-4:** Data collection, preprocessing, and initial feature extraction (e.g., numerical and text-based features).
- **Weeks 5-6:** Clustering algorithm implementation, optimization, and dimensionality reduction for visualization.
- **Weeks 7-8:** LLM integration for cluster interpretation and recommendation generation, plus initial UI development.
- **Weeks 9-10:** System integration, testing (e.g., clustering metrics, LLM output quality), and debugging.
- **Weeks 11-12:** Final refinements, visualization enhancements, documentation, and presentation preparation.

Tools and Technologies

These are suggested tools and technologies; feel free to use alternatives that you find more suitable or are more comfortable with.

- LLM: Grok 3 (via <https://x.ai/api> for higher quotas).
- Clustering and Data Processing: Scikit-learn (clustering, metrics), Pandas, NumPy.
- NLP: Hugging Face Transformers (sentiment analysis), NLTK/Spacy (text pre-processing).
- Visualization: Matplotlib, Seaborn, Plotly for cluster plots; Shiny or Streamlit or Flask for UI.
- Agent Frameworks: LangChain or AutoGen for coordinating preprocessing, clustering, and LLM tasks.

Learning Outcomes

- Technical Skills:
 - Apply unsupervised learning techniques (e.g., clustering) to real-world customer data.
 - Perform data preprocessing, including handling numerical and text data for mixed datasets.
 - Utilize LLM prompt engineering for feature extraction and natural language generation.
 - Evaluate clustering performance using quantitative metrics (e.g., silhouette score) and qualitative business relevance.
 - Develop interactive data visualizations and user interfaces for stakeholder communication.
- Soft Skills:
 - Enhance teamwork through collaborative development of an end-to-end system.
 - Improve project management by meeting milestones and coordinating tasks.
 - Strengthen communication via clear documentation and presentations.
- Ethical and Professional Skills:
 - Address ethical concerns like bias in segmentation or privacy in customer data handling.
 - Develop critical thinking in evaluating model outputs for business applicability.
- Portfolio Impact:
 - Build a marketable project showcasing data science, NLP, and visualization skills with industry relevance.

Resources

Here are a list of resources that you may find as useful starting points for your research.

- Datasets:
 - Customer Segmentation Tutorial (<https://www.kaggle.com/datasets/vjchoudhary7/customer-segmentation-tutorial-in-python>).
 - Mall Customers (<https://www.kaggle.com/datasets/kandij/mall-customers>).
 - Customer Personality Analysis (<https://www.kaggle.com/datasets/imakash3011/customer-personality-analysis>).
- Clustering:
 - Scikit-learn Clustering Documentation (<https://scikit-learn.org/stable/modules/clustering.html>).
- NLP:
 - Hugging Face Transformers (<https://huggingface.co/docs/transformers/>)
 - NLTK (<https://www.nltk.org/>).
- Visualization and UI:
 - Streamlit Docs (<https://docs.streamlit.io/>)
 - Flask Docs (<https://flask.palletsprojects.com/>).
- Agent Frameworks:
 - LangChain (<https://python.langchain.com/docs/>)
 - AutoGen (<https://microsoft.github.io/autogen/>).

Deliverables

For detailed requirements of the project report and presentation, please consult LEARN.

- A working customer segmentation system, hosted locally or on a cloud platform (e.g., ShinyApps.io, Streamlit Sharing, Render), capable of clustering customers and generating segment descriptions and marketing recommendations.
- A final report detailing methodology (preprocessing, clustering, LLM usage), evaluation results (e.g., clustering metrics, recommendation relevance), and challenges.
- A presentation demonstrating the system, showcasing segment visualizations, LLM-generated outputs, and discussing technical and business insights.
- Optional: Interactive visualizations (e.g., 2D cluster plots, segment dashboards) to highlight customer patterns.

Potential Challenges and Mitigations

- **Data Quality:** Incomplete or inconsistent datasets may lead to poor clustering results.
 - Mitigation: Perform thorough preprocessing (e.g., impute missing values, remove outliers) and validate data quality early.
- **Choosing Optimal Clusters:** Determining the right number of clusters (e.g., for K-means) may be difficult.
 - Mitigation: Use elbow plots, silhouette scores, and business context to select the optimal number of clusters.
- **LLM Output Consistency:** LLM-generated segment descriptions or recommendations may vary in quality or tone.
 - Mitigation: Refine prompts, test outputs iteratively, and incorporate human review to ensure alignment with business goals.
- ****Feature Selection**:** Irrelevant or redundant features may degrade clustering performance.
 - Mitigation: Apply feature selection techniques (e.g., correlation analysis, PCA) and validate features with domain knowledge.
- **UI Integration:** Displaying complex clustering results and LLM outputs in a user-friendly interface may be challenging.
 - Mitigation: Use Shiny or Streamlit or Flask for rapid UI development, focus on clear visualizations, and test usability with peers.
- **Ethical Concerns:** Segmentation may inadvertently reinforce biases (e.g., based on demographics) or misuse customer data.
 - Mitigation: Audit clusters for bias, anonymize sensitive data, and document ethical considerations transparently.
- **Team Coordination:** Managing diverse tasks (data, clustering, LLM, UI) across team members may lead to delays.
 - Mitigation: Use GitHub for code sharing, Trello/Slack for task tracking, and assign clear roles and milestones.

Scalability and Extensions

Here are some ideas for stretch goals if time permits.

- Integrate real-time customer data (e.g., via X API or web scraping) to update segments dynamically.
- Add a feedback loop where marketing campaign results refine segment definitions.
- Use Grok 3's voice mode (available on iOS/Android apps) to present segment summaries for stakeholders.
- Incorporate advanced clustering techniques (e.g., Gaussian Mixture Models) or ensemble methods for improved segmentation.

Ethical Considerations

- Ensure customer data privacy by anonymizing personally identifiable information and adhering to data protection guidelines.
- Mitigate biases in segmentation (e.g., avoiding unfair targeting based on age or income) through bias audits and diverse feature selection.
- Maintain transparency by clearly documenting how segments are formed and how recommendations are generated.
- Discuss the societal impact of targeted marketing, balancing business goals with responsible AI practices.

Documentation and Iterative Improvement

Students should maintain detailed documentation of their methodology, challenges, and iterative refinements to demonstrate the development process. Regular updates to the project report and code comments will enhance transparency and showcase learning outcomes.

Summary

This project combines clustering, NLP, and multi-agent systems to create a customer segmentation system with real-world business applications. It offers students a chance to develop technical skills in data science and soft skills in teamwork, while addressing ethical considerations in AI-driven marketing.

Project Code	DataConsulting1
Type	DATA603 (Industry based internship)
Title	Third-Party Data Integration for Predictive Modeling
Industry partner	Data Consulting
Location	University of Canterbury
Students	Group

Project Summary

Background

This project aims to build a structured, queryable data library integrating multiple third-party, publicly available datasets with the existing Human Movement Data (HMD) PostgreSQL/PostGIS framework. The goal is to enable predictive modeling by enriching the HMD with complementary contextual data such as Census, traffic counts, weather, tides, population change, and other environmental or socioeconomic indicators.

The resulting library will be designed for repeatable updates, standardized schemas, and interoperability with analysis workflows in Jupyter/Mito, allowing users to perform joined queries, correlation analyses, and model training seamlessly.

We are currently developing a physical activity strategy for the Bay of Plenty region.

The goal of this strategy is to encourage more people to regularly complete a minimum moderate physical activity (150 mins / week). Current trends are seeing this move away from traditional indoor or structured sport participation to more informal outdoor activities and particularly activities that support activity in natural environments (e.g. native bush, coast, walking, cycling etc).

Problem statement

To design and implement a modular, scalable data library of third-party datasets that complements existing HMD assets, enabling advanced modeling (e.g. visitation prediction, trend analysis, or risk modeling) through structured joins across multiple data sources.

Developing and collating this data to effectively establish a predictive model for future use scenarios.

Scope of work

1. Data Sourcing & Exploration

- Identify and collect publicly available datasets relevant for modeling visitation and behavior patterns:
 - Census data (e.g., Stats NZ or ABS)
 - NZ Health Survey Data for the Bay of Plenty
 - Traffic counts (e.g., NZTA open data portals)
 - Weather and climate data (e.g., NIWA, BOM)
 - Tide data (e.g., LINZ, NOAA)
 - Population movement / shift data
 - Other contextual datasets: land use, socioeconomic indices, etc.
 - Validate licensing, data quality, frequency, and coverage.

2. Schema Design & Standardization

- Develop standardized schemas for each dataset aligned with PostgreSQL best practices.
- Define key fields for joins:
 - Spatial joins: via PostGIS geometry/geography columns
 - Temporal joins: via standardised date, month, year fields

- Key identifiers: area codes (e.g., meshblock, SA2, TA)
 - Document schemas in a shared repository.
- 3. Database Structuring & Integration
 - Implement schemas in the PostgreSQL/PostGIS environment alongside HMD tables.
 - Create ETL scripts (Extract-Transform-Load) for each dataset to:
 - Ingest data
 - Clean and normalize fields
 - Create geometry columns
 - Index key columns for performance
 - Set up version-controlled pipelines for repeatable updates.
- 4. Query & Join Framework
 - Build pre-defined SQL views and join templates for:
 - Linking contextual datasets to HMD visitation tables
 - Aggregating by time, geography, or attribute
 - Feeding downstream Mito/Jupyter notebooks
 - Provide examples in Mito/Jupyter for querying and joining data.
- 5. Predictive Modeling Enablement
 - Develop baseline models that combine HMD data with third-party inputs to predict:
 - Physical activity participation “inside” vs “outside”
 - Visitation volume
 - Dwell time
 - Seasonality
 - Demographic shifts
 - Establish a data-science-ready environment for correlation and regression analyses.

Deliverables

1. Data Library
 - PostgreSQL/PostGIS schemas for each dataset
 - Indexed tables with consistent spatial and temporal keys
 - Documented ETL/update scripts
2. Schema Documentation
 - Schema definitions and field descriptions
 - Join logic and lookup tables
3. Query Templates
 - SQL and Mito-ready notebooks for joins and aggregations
 - Sample analyses linking HMD and contextual datasets
4. Final Report
 - Overview of integrated datasets
 - Data relationships and potential modeling use cases
 - Recommendations for future enrichment and maintenance

Skills required

- Data engineering (PostgreSQL, PostGIS)
- ETL scripting (Python, SQL)
- Schema design and normalization
- Statistical analysis and feature engineering
- Data visualization and reporting (Mito, Jupyter)
- Understanding of spatial and temporal data joins

Project Code	DataConsulting2
Type	DATA603 (Industry based internship)
Title	Enhancing Movement data through the integration of third party head count data for SLSNZ patrolled beaches.
Industry partner	Data Consulting
Location	University of Canterbury
Students	Group

Project Summary

Background

This project aims to provide students with an opportunity to explore advanced data normalisation techniques using real-world mobile location data. Baseline data sets will be used to enhance the validity/ accuracy of the normalisation methodology.

The primary objective of this project is to use third party head count data to normalise mobility movement data. Students will apply Surfzone AI head count data to increase the accuracy of the movement data set, then employ data science techniques to find correlations between SLSNZ patrol headcount data and assess accuracy and assist with normalise movement at 35 + patrol locations around NZ.

Problem statement

Clearly articulate the problem statement, objectives, or research questions...

Scope of work

1. **Understand the White Paper Concepts:** The white paper outlines key challenges and methods for mobile location data normalisation. Students should begin by thoroughly understanding these concepts and identifying which techniques are applicable to the Azira visitation dataset.
2. **Develop a Normalisation Model:** Using the concepts of data cleansing, smoothing, sampling, baselining, and aggregation as described in the white paper, students will design and implement a normalisation model. This model should:
 - Minimise data volatility and eliminate extreme outliers.
 - Create consistent baselines to support time-series analysis.
 - Ensure the normalised dataset reflects realistic and actionable insights.
3. **Validate the Model:** Apply statistical methods to validate the performance of the developed model against known benchmarks. This will involve comparing normalised data with ground truth datasets including SLSNZ patrol headcount and Surfline SurfZone AI data sets.
4. **Extrapolate the Model:** Apply model to other patrol locations where AI headcounts are not available, to improve accuracy of ground truth data sets and movement data sets.
5. **Documentation and Reporting:** Document the model development process, methodologies, and validation results. The report should include visualisations, statistical comparisons, and interpretations of the enhanced data.

Deliverables

- **Normalisation Model:** A model using Python or R, with accompanying code and a Jupyter notebook (or R Markdown file) that explains each step.
- **Validation Report:** A report that includes a performance analysis of the model, visualisations (e.g., trend lines, before-and-after comparisons), and recommendations for future improvements.

Skills required

By the end of this project, students will have hands-on experience building a data normalisation model using complex, real-world datasets. This experience will enhance their understanding of data preprocessing, the challenges of working with mobile location data, and how to produce high-quality data outputs for business or research purposes.

References

- Access to actual movement datasets through Data Consulting and SLSNZ
- Access to Surfzone AI and SLSNZ patrol head count data
- The "Mobile Location Data Normalisation: An Azira Primer" white paper as a foundational reference.
- Support and guidance from Data Consulting and the SLSNZ Coastal Safety team.

Project Code	EECA1
Type	DATA605 (Industry based research project)
Title	Synthetic Smart Meter Data Generation for Residential Energy Modelling
Industry partner	The Energy Efficiency and Conservation Authority (EECA)
Location	University
Students	Could be either an individual or a group

Project Summary

Background

Residential energy end-use datasets do more than record users' aggregate energy consumption. They can implicitly reveal activities occurring within a dwelling, such as appliance use and occupants' daily routines. Consequently, the direct use of such data raises privacy and data ethics concerns that necessarily restrict access and limit analysis. However, the use of these data is essential in energy research and policy. A pragmatic remedy is to generate synthetic data that emulates the original dataset's structure, while severing any direct link to individual households, thereby enabling responsible data sharing and downstream analysis.

Summary

This project involves developing methods for the synthesis and comparison of residential energy end use (smart meter) data across varying temporal and spatial resolutions. The end use dataset held by EECA consists of real, anonymised high-resolution electricity demand records for approximately 50k households across New Zealand. This project aims to include stochastic time-of-use demand data synthesis, utilizing decomposition and/or machine learning methods, and will produce outputs useful for EECA's TIMES-NZ model, among numerous other use cases.

Project Code	EECA2
Type	DATA605 (Industry based research project)
Title	Improving the Regionalisation of New Zealand's Energy End Use Database
Industry partner	The Energy Efficiency and Conservation Authority (EECA)
Location	University
Students	Could be either an individual or a group

Project Summary

Background

The [Energy End Use Database](#) (EEUD) provides detailed estimates of energy consumption across the New Zealand economy, both at national and regional levels. It includes energy demand broken down by end use, fuel, sector, region, and technology. Although this framework already underpins important energy efficiency and emissions modelling, there are opportunities for methodological improvement, particularly in regionalizing energy use, where geospatial and sectoral estimates are derived indirectly via scaling or optimization methods when primary data is not available.

Summary

This project will focus on enhancing or redesigning the regionalisation component of the EEUD methodology. The work will involve reconstructing the existing process to understand its assumptions, limitations, and data dependencies, and developing a workflow that integrates both spatial and statistical methods. Potential improvements including incorporating additional regional energy and labour statistics, enhancing data quality, developing a prototype regionalisation model that improves accuracy, and producing a data processing pipeline with transparent, reproducible code.

Project Code	ERA1
Type	DATA605 (Industry based research project)
Title	Colourful birds
Industry partner	ERA Group
Location	University
Students	Group

Project Summary

About ERA Group

We are a global firm of 1100+ consultants operating in +65 countries, specializing in Cost Intelligence. We play a crucial intermediary role between the Client's Procurement needs and Suppliers in the market. With expert knowledge in multiple categories, we offer specialist knowledge and procurement resources to our clients which they would otherwise struggle to resource on their own.

About Colorful Birds

The Art of Connection

A service for carers who want to re-connect with their children during the supervised visit in a warm environment and like to engage in playful activities that help carers to bond with their children. We are a small team of qualified professionals who provide a service of supervision both in community and venue based.

The Family Court (Ministry of Justice) will order that the estranged parent is allowed to see their child(ren) under supervision.

It is a valuable moment for Colourful Birds to play a role in healing the relationship between the child(ren) and the estranged parent which plays directly into the mental wellbeing of all involved.

Business Problem

The oversight body in New Zealand is ANZASCS responsible for training and certifying Supervised Contact providers.

ANAZASCS has been severely underfunded over the years, and their systems are reflective of this.

Recently, these limitations came to a head when ANZASCS tendered for a Ministry of Justice RFP. This RFP has been postponed for a year giving Supervised Contact Providers and ANZASCS a window of opportunity to professionalize the systems.

If ANZASCS does not win the tender or professionalize their systems, the entire ecosystem of Supervised Contact in NZ is at risk of exposing children to further harm.

Objective

Use natural language and AI to build an end-to-end solution (e.g., Google FireBase, MS CoPilot, etc).

Colourful Birds needs an end-to-end system capable of managing, monitoring and reporting our workload.

We would like to create an app which can be rolled out to other Supervised Contact providers around the country.

ANZASCS in their oversight capacity, needs an interface in this app to monitor and quality check Supervised Contact throughout the country.

Implementation Strategy

1. Create the team and define the roles (spokesperson, project lead, technical lead, business analyst)
2. Create and maintain a project plan: challenges, scope, deliverables, outcomes, workflow, milestones, deliverable dates, RACI (Responsible, Accountable, Consulted, and Informed)
3. Business Analysis: processes, datamodel
4. Identify the infrastructure to be used
5. Create, test, modify, test, modify the app.
6. Launch

Expected Outcome

1. A well-documented and easy to understand and manage application
2. A cost-efficient architecture. Costs should be consumption-based with low- or no- implementation fee

We anticipate the process to include the following:

- End-to-end case handling
- Unified messaging
- Secure, 2FA, complies with the applicable laws and regulations

Future Outcomes

A successful application will have a demand in the market and generate a financial reward.

If ANZASCS mandates the use of our solution, expect the app to be rolled out widely and require ongoing development and maintenance. In such an event, a separate company will be set up and students will have an opportunity to participate. This company will be a start-up allowing students to learn how to develop a company from the ground up.

This will be mentored by Roelant Hofmans (from ERA).

Project Code	ERA2
Type	DATA605 (Industry based research project)
Title	Translation Agent
Industry partner	ERA Group
Location	University
Students	Group

Project Summary

About ERA Group

We are a global firm of 1100+ consultants operating in +65 countries, specializing in Cost Intelligence. We play a crucial intermediary role between the Client's Procurement needs and Suppliers in the market. With expert knowledge in multiple categories, we offer specialist knowledge and procurement resources to our clients which they would otherwise struggle to resource on their own.

Business Problem

We are often faced with having to translate to and from English. Supplier contracts, lease agreements, vendor profiles, our reports, invoices, etc.

The easiest method – and the most error-prone - is to take a subscription to one LLM and translate one-way hoping that the latest version of the LLM does a good job.

For minor languages, the risks are greater simply because the corpus is much smaller.

Recent activities in Vietnam have highlighted these challenges.

Objective

Achieving a translation that carries the same sentiment as the original, has the same intent, and is context-accurate.

Implementation Strategy

1. Create the team and define the roles (spokesperson, project lead, technical lead, business analyst)
2. Create and maintain a project plan: challenges, scope, deliverables, outcomes, workflow, milestones, deliverable dates, RACI (Responsible, Accountable, Consulted, and Informed)
3. Collate specimens to be translated
4. Identify the LLM / LLM's to be used
5. Create User Interface
6. Output the translated document in the same format as the original and include scoring

Expected Outcomes

1. A well-documented and easy to understand process in Jupyter Notebook.
2. A cost-efficient architecture. Costs should be consumption-based with low- or no- implementation fee.

We anticipate the process to include the following:

- Input in source language -> target language -> Output in source language
- Compare and score Input and Output for sentiment, intent, context
- Target and output should emulate the input formatting

Project Code	Erosion
Type	DATA601 (Academic research project)
Title	Improving our Understanding of Wave-Atmosphere Interactions and Drivers of Coastal Erosion from Multiple Datasets
Department	University of Canterbury (Maths and Stats)
Location	University of Canterbury
Students	Group

Project Summary

Overview

Coastal erosion is one of Aotearoa New Zealand's most pressing environmental challenges, with high-energy storm waves being a main driver of erosion. This project will explore how interactions between atmospheric conditions and ocean wave processes contribute to shoreline change by identifying the storm types and conditions which drive the most severe erosion.

Students will utilise new satellite-derived shoreline position data from CoastSat NZ (data source 1), wave buoy observations from the South Island Wave Monitoring Network (data source 5), wave hindcasts (data source 3), and high-resolution atmospheric model outputs for New Zealand (data source 6). A central task will be the integration and wrangling of these datasets, which have different formats, spatial resolutions, and sampling frequencies, to enable robust temporal and spatial analysis.

Objectives of the project can be:

1. To classify storm types based on their synoptic and wave characteristics, and identify which types are associated with more severe coastal erosion. Unsupervised machine learning methods can be used.
2. Another key component involves assessing the accuracy of wave and weather models by comparing their outputs to observed data.
3. Using I and II to inform predictions of shoreline responses to future storm events. Data driven prediction methods can be explored here.

Why this Project?

Climate change is increasing the frequency and severity of coastal hazards and there is a growing need for data-driven tools to support better understand storm types associated with more severe erosion. This will lead to more accurate event forecasting and resilient coastal planning. By consolidating and analysing these datasets, the project will generate important insights into the drivers of coastal change. The results will support improved hazard forecasting and inform ongoing coastal monitoring programmes put in place by coastal managers (e.g. Christchurch City Council and Environment Canterbury). The project outputs will help benchmark future change through a clearer understanding of past processes and provide real-world benefit to coastal managers and communities.

Data sources

Shoreline change:

1. <https://uoa-eresearch.github.io/CoastSat/#11/-43.4113/172.7408>
2. <https://github.com/UoA-eResearch/CoastSat/blob/main/README.MD>

Wave hindcasts:

3. WHACS: Wave Hindcast for the Australian Climate Service (CSIRO Australia) (1979-2025):
<https://data.csiro.au/collection/csiro:64350>

In-Confidence

4. NZ Wave Hindcast (1993-2018):

<https://uoa-eresearch.github.io/waves/hindcast.html>

Wave-buoy data:

5. <https://southislandwaves.co.nz/>

You will also get access to full spectral data from the buoys via a separate online portal with a specific username and password.

Description of Spotter wave buoy parameters:

https://content.sofaroccean.com/hubfs/Spotter%20product%20documentation%20page/Sofar%20-%20Technical_Reference_Manual.pdf

WRF/atmospheric simulation and weather station data:

6. <https://www.envlib.org>

Timeline

- Weeks 1-2: Project planning, dataset selection, refining aims
- Weeks 3-4: Data collation, cleaning, and preprocessing
- Weeks 5-10: Analysis of wave-atmospheric processes and model data
- Weeks 11-12: Final refinements, performance evaluation, report writing, and presentation preparation.

Deliverables

For detailed requirements of the project report and presentation, please consult LEARN.

- Creation of storm typology and analysis of corresponding wave characteristics and typical severity of erosion associated with these different storm types.
- A statistical comparison of model vs observed data accuracy.
- A final report detailing the methodology, results, discussion, and conclusions.

Project Code	Foleys1
Type	DATA603 (Industry based internship)
Title	Inventory pricing and review
Industry partner	Foleys
Location	Christchurch or Dunedin (preferred if possible)
Students	Individual

Project Summary

Background

Foleys have tens of thousands of individual inventory items. Many items have different industry codes but are the same/similar inventory items being procured from multiple suppliers and produced by multiple manufacturers. Our suppliers provide their own price books. Foleys current pricing model is cost plus which means that Foleys make less margin if they procure at a lower price. It also means that customers may have a highly variable price experience based on how effective our procurement is.

Problem statement

There are several potential opportunities for AI-driven initiatives within Foleys' procurement and pricing processes, including:

- **Price book analysis:** Compare merchant price books against internal and external data (e.g., usage, historical sales) to identify market prices, best-buying opportunities, and high-margin product lines.
- **AI-based pricing agents:** Develop AI agents or models capable of making ongoing procurement and pricing recommendations, including automated verification of merchant invoices against published price books.
- **Data consolidation and harmonisation:** Use AI to consolidate and normalise product naming conventions, detect duplicate or equivalent items, and establish consistent item groupings across suppliers and manufacturers.

Scope of work

Develop an AI supported process for performing various analysis and consolidation tasks using our merchant price books, external web-based pricing information and comparing to our actual usage based on our own systems data.

Deliverables

AI tool to perform ongoing data analysis across the three information sources listed above i.e. merchant price books, web based pricing and our own ERP information.

Data / resources provided

- Internal procurement and sales data
- External price books
- Support from Procurement Administrator

Skills required

Required

- Strong data analysis and problem-solving skills.
- Experience with Python and data science libraries (e.g., pandas, NumPy, scikit-learn).

- Familiarity with machine learning and natural language processing (NLP) techniques, particularly for data matching or entity resolution.
- Knowledge of data cleaning, transformation, and integration across multiple structured data sources.

Desired

- Familiarity with web scraping or API-based data retrieval.
- Understanding of AI-assisted decision-making systems or recommender models.

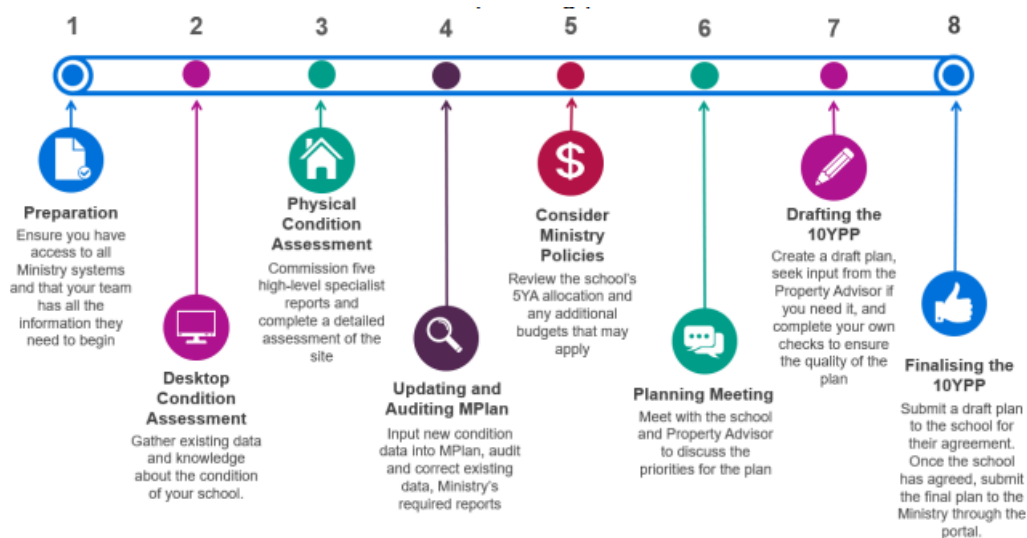
Project Code	Foleys2
Type	DATA603 (Industry based internship)
Title	Building condition assessment report for MoE
Industry partner	Foleys
Location	Christchurch – Office / University
Students	Individual

Project Summary

Background

Foleys is engaged by the Ministry of Education (MoE) to prepare 10-Year Property Plan (10YPP) Physical Condition Assessment reports for schools across New Zealand. These reports cover asset categories such as Electrical, Drainage, Plumbing, Roofing, and Heating systems, forming a critical part of each school's long-term maintenance and capital planning.

Currently, each report requires experts to conduct on-site inspections and document the condition of various assets using photos and structured observations. The information must then be compiled into a formal MoE-compliant report. This process is highly time-consuming and administratively intensive, often requiring many hours of manual consolidation, formatting, and verification per school.



Problem statement

The key challenge lies in efficiently transforming **raw, unstructured site data** (e.g., inspection notes, photos, and asset condition descriptions) into a **fully formatted, compliant MoE report** that meets consultant and regulatory requirements.

Manual report preparation introduces delays, variability, and administrative costs. An AI-supported system could assist in automatically:

- Interpreting and classifying the data captured during site visits.
- Generating structured text sections aligned with MoE templates.
- Embedding relevant photos and formatting outputs according to client and MoE standards.
- Validating that all required sections and compliance elements are complete.

The project seeks to design and test a **proof-of-concept AI model or workflow** that can automate some or all of this process.

Scope of work

Develop an AI supported process for taking the guidance provided by the MoE and the requirements of our consultant customers and the information captured during the site visit to compile a comprehensive report with minimal administrative resource.

Deliverables

AI tool to convert information from site visit into compliant MoE report

Data / resources provided

- MoE guidance resources on 10YPP and specialist reports
- Guidance requirements from our clients
- Internal example reports
- Access to Christchurch based MoE reports team

Skills required

Required

- Proficiency in Python and data processing libraries (e.g., pandas, docx, pdfplumber).
- Understanding of natural language processing (NLP) and text generation methods.
- Familiarity with document automation tools or APIs (e.g., OpenAI, Azure OpenAI, or LangChain for structured output).
- Strong analytical and problem-solving skills, with attention to detail in formatting and compliance.

Desired

- Experience in template-based report generation or data-to-document pipelines.
- Knowledge of regulatory or government reporting frameworks.

References

See above and [10 Year Property Plan information for consultants - Ministry of Education](#)

Project Code	Foleys3
Type	DATA605 (Industry based research project)
Title	Plumbing AI Compliance assistant proof of concept model
Industry partner	Foleys
Location	University
Students	Individual

Project Summary

Background

Foleys operates in a highly regulated industry where adherence to building codes, health and safety standards, and environmental regulations is critical. Ensuring consistent compliance across a wide range of services including residential, commercial, and industrial requires significant administrative effort, ongoing staff training, and real-time awareness of regulatory changes.

As part of a potential commercial initiative, this is an initial exploratory project to support the development of a compliance AI-powered assistant to be used to support field staff in meeting regulatory obligations and industry standards

Problem statement

The AI powered assistant would be designed to enable field staff to ask typical questions such as what is the maximum temperature for domestic hot water system and the AI would reference current plumbing, electrical, and building related regulations from local councils and national authorities when providing an answer.

Scope of work / expected methodology

The student will investigate the feasibility of developing an AI-powered compliance assistant for plumbing and building services. The focus will be on defining the problem space, assessing available regulatory data sources, and developing a proof-of-concept prototype that demonstrates how an AI model can interpret and respond to compliance-related queries.

Key activities will include:

- Conducting a literature and technology review on AI assistants for regulatory or compliance domains.
- Evaluating potential approaches for extracting and structuring regulatory information (e.g., retrieval-augmented generation, semantic search, or rule-based systems).
- Designing and developing a minimal viable model (conceptual or prototype) capable of answering selected compliance questions.

Deliverables include a conceptual or functional prototype, a technical report detailing design and findings, and a presentation summarising the value and feasibility of the solution.

Deliverables

Develop a conceptual model, understanding the feasibility, value and implementation options

Data / resources provided

- Relevant Australian, NZ and regional codes and regulations

Skills required

Required

- Applied data science and machine learning, particularly natural language processing (NLP).
- Strong analytical and written communication skills.

Desired

- Experience with Python and relevant libraries (e.g., LangChain, Hugging Face, OpenAI API, spaCy).
- Familiarity with text data preprocessing, information retrieval, and knowledge-base construction.
- Ability to interpret regulatory or technical documentation and translate it into structured data or rules.

Project Code	FRR1
Type	DATA601 (Academic research project)
Title	Development of a Ground Motion to Intensity Conversion Equation (GMICE) for GeoNet's Felt RAPID report data
Department	School of Mathematics & Statistics
Location	University of Canterbury
Students	Group

Project Summary

Background

Eyewitness accounts of the effects of earthquakes have been used by scientists for hundreds of years. The recent introduction of interactive, online questionnaires (also referred to as felt reports) generates vast quantities of data from members of the public who experience shaking of different intensities during an earthquake. This creates a large sample size immediately available to researchers. These data have the potential to be used for solving problems such as earthquake hazard and seismic risk assessment. These data are known as macroseismic data, as they assess the overall effects of an earthquake across an area, whereas individual instrumental measurements represent only ground motions at that specific site.

In Aotearoa New Zealand, felt reports are considered a valuable way of obtaining information on how people across the country have experienced an earthquake. Online data collection began via GeoNet's website in 2004, including questions specifically aligned with the intensity thresholds in the Modified Mercalli Intensity Scale.

Following the Christchurch earthquakes in 2010 and 2011, it became apparent that the felt reports for damaging shaking (of most interest to researchers) were only sparsely reported. There was a concurrent growing demand from the public for greater user-experience in terms of products and services being available and responsive at any time. Based on public feedback, the Felt RAPID (FR) survey was implemented. It represented a conceptual shift to an easy-to-use, cartoon-based survey. Its main purpose is the immediate assessment of an earthquake's impact, i.e. whether it is severe or not, before other sources of information are available.

Problem statement

This project aims to develop a Ground Motion Intensity Conversion Equation which permits the conversion of macroseismic intensity measurements into units generated by scientific instrument (Peak Ground Acceleration/Peak Ground Velocity).

The project will address several the critical research questions:

1. What is the most effective way to model the relationship between the MMI scale values and pga/pgv?
2. Which confounding variables associated with earthquakes and their perceived intensity have a significant influence on this relationship?

Expected methodology

Students will engage in accessing and preprocessing data from various sources and assessing the reliability of data through exploratory (spatial) data analysis. They will then explore the relationship between the outcome and the predictor(s). A key focus will be on comparing multiple modelling approaches and, through careful evaluation, identify the most effective way to model this relationship while maintaining interpretability and parsimony.

Deliverables

Best fit model of relationship between macroseismic intensity and scientific measurements (pga and/or pgv).

Assessment of influence of identified variables.

Assessment of limitations of modelling and any unexplored factors.

Data / resources provided

- <https://felt.geonet.org.nz/>
- <https://www.geonet.org.nz/data/types/felt>
- <https://www.geonet.org.nz/earthquake/intensity>
- <https://www.geonet.org.nz/earthquake/mmi>
- <https://quakesearch.geonet.org.nz/>
- <https://www.geonet.org.nz/earthquake/felt/2025p672513> - example earthquake page
- <https://www.geonet.org.nz/data/network/sensor/search> - strong motion stations
- https://www.seismosoc.org/wp-content/uploads/2018/06/SSA_2018_Poster_Final.pdf

Skills required

Knowledge and understanding of a range of modelling techniques, their data requirements and limitations.

Some spatial data science knowledge/experience would be an advantage.

References

Haubrock, S., Little, C., McBride, S., & Balfour, N. (2017, 9-12 May). *Keeping up with the citizens – collecting earthquake observations in New Zealand* 20th AGILE Conference on Geographic Information Science, Wageningen, The Netherlands.

Project Code	FRR2
Type	DATA601 (Academic research project)
Title	Development of Intensity Prediction/Attenuation Equation for GeoNet's Felt RAPID Report (FRR) data
Department	School of Mathematics & Statistics
Location	University of Canterbury
Students	Group

Project Summary

Background

Eyewitness accounts of the effects of earthquakes have been used by scientists for hundreds of years. The recent introduction of interactive, online questionnaires (also referred to as felt reports) generates vast quantities of data from members of the public who experience shaking of different intensities during an earthquake. This creates a large sample size immediately available to researchers. These data have the potential to be used for solving problems such as earthquake hazard and seismic risk assessment. These data are known as macroseismic data, as they assess the overall effects of an earthquake across an area, whereas individual instrumental measurements represent only ground motions at that specific site.

In Aotearoa New Zealand, felt reports are considered a valuable way of obtaining information on how people across the country have experienced an earthquake. Online data collection began via GeoNet's website in 2004, including questions specifically aligned with the intensity thresholds in the Modified Mercalli Intensity Scale.

Following the Christchurch earthquakes in 2010 and 2011, it became apparent that the felt reports for damaging shaking (of most interest to researchers) were only sparsely reported. There was a concurrent growing demand from the public for greater user-experience in terms of products and services being available and responsive at any time. Based on public feedback, the Felt RAPID (FR) survey was implemented. It represented a conceptual shift to an easy-to-use, cartoon-based survey. Its main purpose is the immediate assessment of an earthquake's impact, i.e. whether it is severe or not, before other sources of information are available.

Problem statement

This project aims to develop an intensity attenuation relation model for New Zealand based on FRR data.

The project will address several the critical research questions:

1. What is the most effective way to model the relationship between MMI values and the distance from the earthquake hypocentre?
2. Which confounding variables associated with earthquakes and their perceived intensity have a significant influence on this relationship?

Expected methodology

Students will engage in accessing and preprocessing data from various sources and assessing the reliability of data through exploratory (spatial) data analysis. They will then explore the relationship between the outcome and the predictor(s). A key focus will be on comparing multiple modelling approaches and, through careful evaluation, identify the most effective way to model this relationship while maintaining interpretability and parsimony.

Deliverables

Best fit model of relationship between macroseismic intensity and distance from the earthquake hypocentre.

Assessment of influence of identified variables.

Assessment of limitations of modelling and any unexplored factors.

Data / resources provided

- <https://felt.geonet.org.nz/>
- <https://www.geonet.org.nz/data/types/felt>
- <https://www.geonet.org.nz/earthquake/intensity>
- <https://www.geonet.org.nz/earthquake/mmi>
- <https://quakesearch.geonet.org.nz/>
- <https://www.geonet.org.nz/earthquake/felt/2025p672513> - example earthquake page
- <https://www.geonet.org.nz/data/network/sensor/search> - strong motion stations
- https://www.seismosoc.org/wp-content/uploads/2018/06/SSA_2018_Poster_Final.pdf

Skills required

Knowledge and understanding of a range of modelling techniques, their data requirements and limitations.

Some spatial data science knowledge/experience would be an advantage.

References

Haubrock, S., Little, C., McBride, S., & Balfour, N. (2017, 9-12 May). *Keeping up with the citizens – collecting earthquake observations in New Zealand* 20th AGILE Conference on Geographic Information Science, Wageningen, The Netherlands.

Project Code	FRR3
Type	DATA601 (Academic research project)
Title	Using isoseismal maps generated from GeoNet Felt RAPID Report (FRR) data to estimate the magnitude and location of an earthquake.
Department	School of Mathematics & Statistics
Location	University of Canterbury
Students	Group

Project Summary

Background

Eyewitness accounts of the effects of earthquakes have been used by scientists for hundreds of years. The recent introduction of interactive, online questionnaires (also referred to as felt reports) generates vast quantities of data from members of the public who experience shaking of different intensities during an earthquake. This creates a large sample size immediately available to researchers. These data have the potential to be used for solving problems such as earthquake hazard and seismic risk assessment. These data are known as macroseismic data, as they assess the overall effects of an earthquake across an area, whereas individual instrumental measurements represent only ground motions at that specific site.

In Aotearoa New Zealand, felt reports are considered a valuable way of obtaining information on how people across the country have experienced an earthquake. Online data collection began via GeoNet's website in 2004, including questions specifically aligned with the intensity thresholds in the Modified Mercalli Intensity Scale.

Following the Christchurch earthquakes in 2010 and 2011, it became apparent that the felt reports for damaging shaking (of most interest to researchers) were only sparsely reported. There was a concurrent growing demand from the public for greater user-experience in terms of products and services being available and responsive at any time. Based on public feedback, the Felt RAPID (FR) survey was implemented. It represented a conceptual shift to an easy-to-use, cartoon-based survey. Its main purpose is the immediate assessment of an earthquake's impact, i.e. whether it is severe or not, before other sources of information are available.

Problem statement

This project aims to use geospatial modelling approaches to create isoseismal maps from FRRs and use these to estimate the location and magnitude of an earthquake.

The project will address several the critical research questions:

1. Which is the most effective way to infer the spatial distribution of different intensity values (MMI) caused by an earthquake?
2. What is the most effective way to model the relationship between the resulting distribution and the magnitude and location of the earthquake?
3. Which confounding variables associated with earthquakes and their perceived intensity have a significant influence on this relationship?

Expected methodology

Students will engage in accessing and preprocessing data from various sources and assessing the reliability of data through exploratory (spatial) data analysis. They will then explore the relationship between the outcome and the predictor(s). A key focus will be on comparing spatial modelling approaches and, through careful evaluation, identify the most effective way to model this relationship while maintaining interpretability and

parsimony. The resulting models will then be used to predict both the location and magnitude of earthquakes and evaluate their performance.

Deliverables

Best fit model to infer the spatial distribution of FRR macroseismic intensity data.

Best fit modelling process to infer the epicentre and magnitude of the earthquake.

Assessment of influence of identified variables.

Assessment of limitations of modelling and any unexplored factors.

Data / resources provided

- <https://felt.geonet.org.nz/>
- <https://www.geonet.org.nz/data/types/felt>
- <https://www.geonet.org.nz/earthquake/intensity>
- <https://www.geonet.org.nz/earthquake/mmi>
- <https://quakesearch.geonet.org.nz/>
- <https://www.geonet.org.nz/earthquake/felt/2025p672513> - example earthquake page
- <https://www.geonet.org.nz/data/network/sensor/search> - strong motion stations
- <https://www.sciencedirect.com/topics/earth-and-planetary-sciences/isoseismal-map>
- https://en.wikipedia.org/wiki/Isoseismal_map

Skills required

Knowledge and understanding of a range of spatial data science modelling techniques, their data requirements and limitations.

References

Haubrock, S., Little, C., McBride, S., & Balfour, N. (2017, 9-12 May). *Keeping up with the citizens – collecting earthquake observations in New Zealand* 20th AGILE Conference on Geographic Information Science, Wageningen, The Netherlands.

Project Code	GWC
Type	DATA605 (Industry based research project)
Title	Forecasting and Managing Regalia Hire Demand
Industry partner	GWC Regalia Hire
Location	University of Canterbury
Students	Could be either an individual or a group

Project Summary

Background

GWC Regalia Hire provides academic dress hire for the majority of tertiary institutions across Greater Canterbury. Based at the University of Canterbury, GWC supplies regalia for graduation ceremonies at UC and other institutions, including gowns, hoods, and caps in a wide range of sizes and institutional colours.

The business plays an essential role in ensuring that thousands of graduates each year have access to correctly sized regalia during peak graduation periods. Their operations require careful coordination of stock, logistics, cleaning, and ceremony scheduling.

Maintaining sufficient stock to meet annual graduation demand is critical for smooth service delivery. However, stocking regalia involves long lead times for manufacture and import, as well as significant storage costs and logistical complexity.

Problem statement

GWC Regalia Hire must maintain enough regalia stock to meet varying graduation cohort sizes and body-size distributions each year. The company currently collects size information only once graduates register to attend ceremonies, typically within a few months of the event, while procurement and manufacturing can require up to a year of lead time.

This results in uncertainty around:

- **Stock sufficiency:** risk of running short of certain sizes and having to borrow stock from Otago
- **Overstocking:** tying up capital and limited storage space.
- **Changing cohort demographics:** year-to-year variation in graduate numbers, institutions served, and size distributions.
- **Operational scheduling:** multiple ceremonies and parades held within compressed timeframes increase turnaround pressure and can increase stock requirements.

Developing a robust, data-driven forecast model for regalia demand will allow GWC Regalia Hire to plan inventory, reduce risk, and improve efficiency.

Scope of work / expected methodology

The project will analyse historical hire data and develop forecasting tools to optimise stock management and procurement decisions.

1. Historical data extraction and analysis
 - Extract regalia hire records over the last 18 months (new system) to 5 years (old system).
 - Identify seasonal, institutional, and event-specific demand patterns.
 - Quantify size distributions and changes over time.
2. Size requirement modelling
 - Analyse how height and body-size distributions vary by year, qualification, and institution.
 - Define “worst-case” demand scenarios for each regalia size class.

3. Operational dynamics review
 - Examine the structure of ceremonies, parades, and hire/return periods.
 - Assess how ceremony spacing and turnaround windows affect total stock requirements.
4. Forecasting and predictive modelling
 - Build models to predict future stock requirements based on early-available information (e.g. enrolment data, qualification completions, historic trends).
 - Evaluate model accuracy and incorporate uncertainty ranges for procurement planning.

Deliverables

- **Comprehensive report** covering data analysis, model results, and operational insights for each scope component.
- **Interactive forecasting tool** (e.g. spreadsheet, dashboard, or Python/R application) enabling GWC staff to update inputs annually and forecast demand by size, institution, and ceremony.
- **Documentation and user guide** describing data requirements, update procedures, and interpretation of outputs.
- **Recommendations** for data collection improvements (e.g. early surveying, integration with graduation registration systems).

Data / resources provided

- Historical regalia hire records (transaction logs, size distributions, ceremony schedules).
- Access to GWC Regalia Hire management and operational staff for data context and validation.
- Information on graduation schedules (e.g. UC, Ara Institute, Lincoln University).

Skills required

- Data wrangling and analysis (Python, R, Excel Power Query, SQL or equivalent).
- Statistical or machine-learning forecasting (time-series, regression, or simulation methods).
- Data visualisation and dashboard development (Tableau, Power BI, Plotly Dash, etc.).
- Strong communication and report-writing skills to translate analysis into actionable recommendations.
- Interest in operations analytics, logistics, or demand forecasting.

References

- [GWC Regalia Hire](#)

Project Code	Headway1
Type	DATA603 (Industry based internship)
Title	Property Market Index
Industry partner	Headway Systems
Location	University – 1 day in Office
Students	Individual

Project Summary

Background

Headway Systems Ltd provides the software product Valbiz to property valuers across the country. Additionally, it is a bulk data supplier to several customers who use it for a variety of property-related applications.

Headway has recently begun offering additional data fields produced in-house, including property estimates. Headway's automated valuation mechanism (AVM) requires a large quantity of sales data to produce accurate estimates. To ensure sufficient volumes, the AVM often requires sales data dating back ~8 months from the present.

The property market can shift significantly over an 8 month period, so Headway uses a region-specific property market index to ensure any older sale prices may reflect the current market. In simple terms, if the index has increased 5% from 8 months ago, then any sales from 8 months ago subsequently have their price increased by 5% - this 'shifts' the sale into the current market.

Problem statement

The current index uses the monthly median sale price for each region as its basis, with a regression model applying a line of best fit. However, the median sale price can be too easily influenced by low sale volumes – especially in smaller regions.

Headway is looking for a new technique to indicate the market volatility that is not solely reliant on the median sale price.

Scope of work

Data Cleaning

- Removing any incomplete sales (e.g., no sale price provided).
- Filtering for the correct sale type – details will be provided.

Pre-Processing

- Produce additional independent variables that may be indicative of market activity, e.g., sale volumes.

Modelling

- A model that produces a numerical index that can be used in adjusting past sale prices.
- Test different indices using Headway's AVM and comparing estimates to actual sale prices.

Deliverables

A new property market index that improves the accuracy of Headway's AVM.

Data / resources provided

- All necessary data
- Access to Headway's AVM for testing any results

- Access to any required domain knowledge

Skills required

- Data cleaning and preprocessing skills
- Machine learning experience using Python and/or R

Project Code	Headway2
Type	DATA603 (Industry based internship)
Title	Dashboard for Sales Volumes
Industry partner	Headway Systems
Location	University – 1 day in Office
Students	Individual

Project Summary

Background

Headway Systems Ltd provides the software product Valbiz to property valuers across the country. This is used as a tool by many for valuation reports and to compile historic notes or data specific to each user.

Headway receives sales data from all territorial land authorities (TLAs) across New Zealand, with the data either sourced directly from councils or through a valuation service provider (VSP). With incoming data obtained from numerous sources, Headway faces a variety of challenges around data quality.

Problem statement

Currently, Headway's processes report sale volumes based solely on file-level counts. However, because some VSPs provide data covering multiple TLAs, these counts are often redundant, and the actual number of sales per region is unclear. Without accurate counts, potential source errors can go unnoticed until customers raise concerns.

Headway aims to address this by developing a dashboard that highlights lower sale volumes and enables us to proactively flag issues to our sources before they reach our clients.

Scope of work

Data Cleaning

- Checking for any incomplete sales (e.g., no sale price provided) and remove from any counts.
- Differentiate between a new sale and an amendment, to avoid a single sale being counted several times.

Power BI or Tableau Dashboard

- A dashboard able to clearly display data for 50+ TLAs – with troublesome TLAs being at the forefront of any visualisations.
- Ability to independently review any TLA and receive more detailed information or visualisations regarding past and current sales volumes.

Deliverables

An accurate and concise dashboard that can highlight regions with lower than usual sale volumes

Data / resources provided

- All necessary sales data
- Access to any required domain knowledge

Skills required

- Data cleaning and preprocessing skills
- Experience with either Tableau or Power BI.

Project Code	Kairos
Type	DATA603 (Industry based internship)
Title	Conversational & Multi-Agent Assistant for a Utility Provider
Industry partner	Kairos Enterprises
Location	University
Students	Could be either an individual or a group

Project Summary

Background

This proposal outlines a consolidated MVP for a multi-modal (chat and voice) conversational assistant tailored for a utility provider. The assistant leverages NLP and multi-agent orchestration to support both public and signed-in customer journeys. It enables seamless transitions between general enquiries and private actions, with capabilities for intent recognition, specialist routing, and support for pre-sales and post sales tasks.

Scope of work

The project is broken down into multiple stages which are outlined below.

Stage 1: Public vs Private Access & Fair-Use Controls

Establish boundaries between public and signed-in interactions, ensuring fair access and privacy. Public enquiries (e.g., pricing, plan comparisons) are protected by fair-use controls, while private actions require sign-in and clear confirmations. Multi-agent design enables seamless coordination without exposing complexity.

Deliverables

- Public enquiry module with fair-use simulation
- Private action module with mock sign-in and confirmation flows
- Multi-agent orchestration framework

Stage 2: Reception & Intent Routing

Create a welcoming entry point that understands customer needs and routes them appropriately. NLPbased intent recognition in chat and voice enables seamless routing to specialist agents with clarifying questions and warm handovers when needed.

Deliverables

- Intent recognition engine using NLP
- Routing logic to specialist modules
- Fallback and clarification handling

Stage 3: Pre-Sales Support (Pricing, Plans & Site Feasibility)

Provide clear, conversational guidance for prospective customers. Support pricing and plan questions, feasibility checks, and alternative suggestions using multi-agent coordination.

Deliverables

- Pricing and plan comparison module
- Feasibility check simulation
- Alternative suggestion logic

Phase 4: Post-Sales Self-Service (Account & Billing)

Enable signed-in customers to manage their accounts via chat or voice. Support billing, updates, and help with requests with clear confirmations and summaries of changes.

Deliverables

- Account and billing self-service module
- Confirmation and summary flows
- Sensitive action safeguards

Expected outcomes

- Conversational Assistant MVP with chat and voice support
- Multi-agent orchestration for specialist routing
- Intent recognition and routing engine
- Pre-sales and post-sales conversational modules
- Fair-use and privacy control mechanisms
- Voice experiences enhancements

Data / resources provided

Support will be provided by Kairos Enterprises throughout the summer.

Project Code	LLM
Type	DATA601 (Academic research project)
Title	Building and fine-tuning a custom large language model using open source GPT frameworks such as NanoGPT or Hugging Face Transformers
Department	School of Mathematics & Statistics
Location	University of Canterbury
Students	Can be an individual or a group

Project Summary

Background

Large language models (LLMs) such as GPT-4 have revolutionised natural language processing by demonstrating powerful capabilities in text generation, summarisation, and reasoning. While large-scale systems are trained on enormous datasets using supercomputing resources, open-source projects such as NanoGPT and the Hugging Face Transformers library make it possible to understand, build, and fine-tune smaller-scale GPT models on standard research or cloud hardware.

This project allows students to explore the practical and conceptual foundations of LLMs by training or fine-tuning a transformer-based model on a chosen text dataset. Students will gain hands-on experience in data collection, cleaning, model configuration, training, and evaluation — the same key processes used in modern AI development pipelines.

Problem statement

Although state-of-the-art LLMs are typically closed and resource-intensive, the underlying transformer architecture can be studied and adapted for smaller, domain-specific applications.

The project challenges students to:

- Collect and prepare a suitable text dataset for model training.
- Train or fine-tune a small GPT-style model.
- Evaluate how data quality, model size, and fine-tuning parameters affect output quality and generalisation.

By the end of the project, students should understand how generative language models work internally and how small-scale fine-tuning can adapt them for specific tasks or domains.

Expected methodology

Students will undertake the following key activities:

1. Framework setup:

- Choose between [NanoGPT](#) or [nanochat](#) (for a minimal, from-scratch implementation) or Hugging Face Transformers (for practical fine-tuning of pre-trained models such as GPT-2, DistilGPT-2, or GPT-Neo).
- Install and configure the required environment locally or on a GPU-enabled cloud instance.

2. Data acquisition and preparation:

- Identify or create a domain-specific text corpus (e.g. news, research abstracts, literature, or open dialogue).
- Clean and preprocess text (tokenisation, normalisation, filtering).

3. Model training and fine-tuning:

- Train a small-scale GPT-style model using either NanoGPT or Hugging Face Transformers.
- Experiment with hyperparameters such as learning rate, batch size, and context length.

- (Optional) Apply parameter-efficient fine-tuning (LoRA / PEFT) to reduce compute cost.

4. Evaluation and analysis:

- Evaluate model outputs using qualitative and quantitative measures.
- Compare the base pre-trained model's responses to those of the fine-tuned version.
- Analyse limitations and propose improvements or future directions.

Deliverables

- Cleaned and documented text dataset.
- Trained or fine-tuned model (and reproducible training script / notebook).
- Report covering data preparation, model training, evaluation, and findings.
- Presentation summarising key results and lessons learned.
- Demonstration notebook or web interface showcasing text generation results (optional)

Data / resources provided

- Publicly available text datasets from sources such as [Project Gutenberg](#), [Hugging Face Datasets](#), or [Kaggle NLP Datasets](#).
- Relevant open-source model repositories and documentation:
 - [NanoGPT](#)
 - [nanochat](#)
 - [Hugging Face Transformers](#)
 - [PEFT / LoRA fine-tuning tools](#)
- Optional access to UC-provided or cloud-based GPU resources.
- Supervisor support for dataset selection, environment setup, and evaluation guidance.

Skills required

- Basic proficiency in Python programming.
- Familiarity with deep learning frameworks such as PyTorch.
- Understanding of text preprocessing and tokenisation.
- Interest in language modelling, AI ethics, and machine learning engineering.

References

Karpathy, A. (2023). NanoGPT: The simplest, fastest repository for training/finetuning medium-sized GPTs. GitHub

Karpathy, A. (2023). Let's Build GPT from Scratch. YouTube Lecture

Hugging Face. (2024). Transformers Documentation. <https://huggingface.co/docs/transformers>

Hugging Face. (2024). PEFT / LoRA Documentation. <https://huggingface.co/docs/peft>

KnowledgeHut (2024). Datasets for Natural Language Processing. <https://www.knowledgehut.com/blog/data-science/data-science-datasets>

Project Code	LLM-SQL
Type	DATA601 (Academic research project)
Title	Develop a LLM Agent to Query SQL Databases
Department	Maths and Stats
Academic supervisor	Phil Davies (Philip.Davies@canterbury.ac.nz)
Location	University of Canterbury
Students	Group

Project Summary

Project Overview

The goal of this project is to develop an advanced AI-powered query system that leverages the capabilities of large language models (LLMs) to interact with SQL databases. Using LangChain, for example, a framework for developing applications powered by language models, your team will create an intelligent agent capable of understanding natural language queries, translating them into SQL commands, executing these commands against a database, and returning the results to the user in a human-readable format.

Objectives

- **Familiarization with LangChain:** Understand the architecture and capabilities of LangChain and how it can be used to build applications with language models.
- **Database Design and Management:** Design and implement a SQL database suitable for storing and retrieving data relevant to the project's focus area.
- **Natural Language Processing:** Develop the LLM agent to accurately interpret and translate natural language queries into SQL commands.
- **Query Execution and Result Formatting:** Ensure the agent can execute SQL queries and format the results in a user-friendly manner.
- **User Interface Development:** Create a simple, intuitive interface for users to interact with the LLM agent.

Deliverables

- **Project Plan:** A detailed project plan outlining the steps, timeline, and resources required.
- **Database Schema:** A well-designed schema for the SQL database, including tables, relationships, and sample data.
- **LLM Agent:** A functional LLM agent using LangChain capable of understanding and processing natural language queries.
- **Interface:** A user-friendly interface for interacting with the LLM agent.
- **Documentation:** Comprehensive documentation covering the development process, system architecture, usage instructions, and future work recommendations (project report).
- **Presentation:** A final presentation demonstrating the project, key findings, challenges encountered, and solutions implemented.

Skills and knowledge required

- Proficiency in Python programming
- Understanding of SQL and database management
- Knowledge of natural language processing (NLP) techniques

- Familiarity with machine learning concepts and frameworks
- Experience with front-end/user-interface development (optional but beneficial)

Tools and technology

- LangChain (or equivalent): Framework for developing applications with language models
- SQL Database: MySQL, PostgreSQL, or SQLite
- Python Libraries: SQLAlchemy, Pandas, NLTK, or SpaCy
- Front-End Framework: Shiny, Streamlit, Flask or Django for creating the user interface (optional)

Assessment criteria

- **Technical Implementation:** Accuracy and efficiency of the LLM agent in processing queries and interacting with the database.
- **Innovation:** Creativity in problem-solving and the application of advanced NLP techniques.
- **User Experience:** Usability and design of the user interface.
- **Documentation and Presentation:** Clarity, thoroughness, and professionalism in documentation and presentation.

Guidelines for writing the report and presentation may be found on the DATA60x LEARN page. Be sure to document all the effort you have made.

Project Timeline

The following is a suggested project timeline.

- Weeks 1-2: Project planning, research, and familiarization with LangChain and relevant tools.
- Weeks 3-4: Database design and initial setup.
- Weeks 5-7: Development of the LLM agent.
- Weeks 8-9: Interface development and integration.
- Weeks 10-11: Testing, debugging, and optimization.
- Week 12: Final documentation, presentation preparation, and project submission.

By the end of this project, your team will have gained practical experience in applying language models to real-world problems, enhancing your skills in natural language processing, database management, and application development.

Learning Outcomes

Technical Skills:

- Weeks 1-2: Project planning, research, and familiarization with LangChain and relevant tools.
- Master data pre-processing techniques for text data, including cleaning, tokenization, and labeling for NLP tasks.
- Develop proficiency in LLM prompt engineering and fine-tuning for tasks like ticket classification, sentiment analysis, and response generation.
- Gain experience in designing and coordinating multi-agent systems using frameworks like LangChain or AutoGen.
- Learn to implement and evaluate NLP models using metrics such as classification accuracy, F1-score, and BLEU/ROUGE for response quality.
- Build practical skills in creating user-friendly interfaces with tools like Streamlit, Flask, or Shiny for data-driven applications.

Soft Skills:

- Enhance teamwork and collaboration through role-based contributions and integrated system development.

- Strengthen project management abilities by adhering to milestones, managing timelines, and coordinating tasks.
- Improve communication skills through clear documentation, presentations, and discussions of technical and ethical considerations.

Portfolio Impact:

- Create a showcase project demonstrating end-to-end data science skills, from data pre-processing to agent-based automation and UI deployment, with strong industry relevance.

Documentation and Iterative Improvements

Students should document challenges, solutions, and iterative refinements to demonstrate the development process clearly, enhancing learning outcomes and project transparency.

Summary

This project balances technical complexity with practical utility, showcasing the power of LLMs and agent-based workflows in data science.

Project Code	MeridianEnergy1
Type	DATA603 (Industry based internship)
Title	Price prediction and machine learning
Industry partner	Meridian Energy
Location	Meridian Energy (Christchurch Office)
Students	Individual

Project Summary

Background

Meridian Energy is New Zealand’s largest renewable electricity generator, producing power from 100% renewable sources—hydro, wind, and solar. As a purpose-driven organisation, Meridian is committed to helping Aotearoa decarbonise and achieve a net-zero carbon future.

This internship is part of Meridian’s DigiGEN (digital Generation) initiative.

Problem statement

Electricity price forecasting is a complex challenge influenced by market dynamics, generation capacity, demand, and external factors. Meridian seeks to explore and compare machine learning approaches for price prediction, evaluating their performance and suitability for operational use.

Scope of work

- Conduct exploratory data analysis (EDA) on historical price and generation data.
- Implement and compare multiple machine learning models (TBD).
- Evaluate model performance and interpretability.
- Document pros and cons of each approach.

Deliverables

Comparative analysis of machine learning models for price prediction at various forecast horizons (next trading period, 4 hours from now etc).

- Performance metrics and visualisations.
- Technical report summarising findings and recommendations.
- Presentation to Meridian.

Data / resources provided

- Access to relevant datasets from Meridian’s generation and market systems.
- Support from Meridian’s generation and ICT teams.
- Inclusion in agile ceremonies and team stand-ups (as part of the Outage Valuation squad).
- Laptop and workspace at Meridian’s Christchurch office.
- Fortnightly discussions with UC academic supervisors and Meridian.

Skills required

- Ability to work independently and communicate findings clearly.
- Curiosity and willingness to engage with domain experts.

Project Code	MeridianEnergy2
Type	DATA603 (Industry based internship)
Title	Accessing new data sources
Industry partner	Meridian Energy
Location	Meridian Energy (Christchurch Office)
Students	Individual

Project Summary

Background

Meridian Energy is New Zealand’s largest renewable electricity generator, producing power from 100% renewable sources—hydro, wind, and solar. As a purpose-driven organisation, Meridian is committed to helping Aotearoa decarbonise and achieve a net-zero carbon future.

This internship is part of Meridian’s DigiGEN (digital Generation) initiative.

Problem statement

Meridian ingests and transforms a variety of different data sources, many of which are shared by our competitors. New data sources offer potential for Meridian to improve the insights, analytics and reporting in ways that provide advantages to Meridian.

This project aims to prototype how to extract a novel (to Meridian) third party data source into a format where Meridian can gain insights into the value of that data source.

Scope of work

- Scope out one to three potential data sources (Meridian and UC to consider – Solcast API is an option)
- Design and prototype a local data pipeline using best practices (e.g., ETL, data validation, versioning).
- Deploy the pipelines to the cloud using a separate Databricks workspace (student/community edition, separate from Meridian infra)
- Provide recommendations for scalable data engineering practices.

Deliverables

- Prototype data pipeline with documentation.
- Data quality and validation framework.
- Technical report outlining architecture, tools, and future recommendations.
- Presentation to Meridian.

Data / resources provided

- Access to relevant datasets from Meridian’s generation and market systems.
- Support from Meridian’s generation and ICT teams.
- Inclusion in agile ceremonies and team stand-ups (as part of the Outage Valuation squad).
- Laptop and workspace at Meridian’s Christchurch office.
- Fortnightly discussions with UC academic supervisors and Meridian.

Skills required

- Ability to work independently and communicate findings clearly.
- Curiosity and willingness to engage with domain experts.

Project Code	NZTA
Type	DATA603 (Industry based internship)
Title	Improved criticality model for transportation networks
Industry partner	NZTA
Location	Christchurch
Students	Individual

Project Summary

Background

Desire to ensure Value for Money from all investment in State Highways. We are looking to develop a network-wide model that helps to identify the best places for us to invest in improvement. The tool is currently in development and has a number of data sources that underpin it. We require greater sophistication in several areas, including most notably how traffic diverts across the network when there is disruption.

Problem statement

The Transport Agency needs to be selective in where it invests its limited funding. Investing in the right places necessitates, amongst other things, an understanding of how network unavailability will impact our customers.

Scope of work

What traffic impacts arise when a section of State Highway becomes unavailable? Specifically, where will our customers reroute, and how will the roads they reroute to manage with the increase in traffic flows (i.e. will rerouted traffic encounter or cause congestion).

Deliverables

Student - Confirm the detour routes data set (where will SH traffic diverts). Note that different Origin Destination pairs will use different detour options (so this may require more complex assessment).

NZTA – total traffic on diverted route (Local + SH)

Student - Determine whether diverted traffic will cause congestion on the diversion route, and related operating speed impacts. This requires 'demographics' of existing route (i.e. theoretical capacity - what's the lowest capacity infrastructure piece - e.g. one lane bridge). Determine what resultant speeds are likely to be achieved based on the ratio of flow to capacity (and/or posted speed limited).

NZTA – Determine Travel Times and Customer TT costs to estimate trip detour and loss rates

The primary output will need to be a GIS Based dataset that can be incorporated into our systems.

Data / resources provided

Data we hold includes - road network maps, traffic volumes, theoretical road capacity thresholds. Asset based restricting e.g. Bridges and speed.

Skills required

Data modelling, Algorithm development, and an interest in transport networks

Project Code	oVRcome1
Type	DATA605 (Industry based research project)
Title	Analyse PPG heart signal data for signs of anxiety
Industry partner	oVRcome
Location	University of Canterbury
Students	Could be either an individual or a group

Project Summary

Background

oVRcome (<https://www.ovrcome.io/>) uses VR to help people overcome mental health conditions such as phobias and social anxiety. An important part of that is determining how anxious people are during our VR therapy so we can tailor the content correctly to their needs. The most common way is to ask them for their anxiety or SUDS (Subjective Units of Distress) score during or immediately after exposure therapy. We desire less intrusive and more real time indicators of such distress. The heart rate signal, specifically Heart Rate Variability (HRV) is a way to do that, This signal is measured by PPG with a sensor on the skin. Most smartwatches do this.

Summary

The goal of this project is to determine what can be inferred from this signal. Firstly recreate existing results using deep learning and data science, then extend such results to the oVRcome dataset. For example, it is claimed that breathing can be detected by the PPG signal. We would like to verify this claim.

Additionally, the heart is believed to respond to a suddenly stressful situation in a consistent but subtle pattern during the 30 seconds following such an incident. Investigating and reproducing such a pattern is also included in this project if the situation permits. Finally, any similar results or techniques that come to attention should be considered potential work.

Project Code	oVRcome2
Type	DATA605 (Industry based research project)
Title	GenAI video creation for VR exposure
Industry partner	oVRcome
Location	University of Canterbury
Students	Could be either an individual or a group

Project Summary

Background

oVRcome (<https://www.ovrcome.io/>) uses VR to help people overcome mental health conditions such as phobias and social anxiety. An important part of that is determining how anxious people are during our VR therapy so we can tailor the content correctly to their needs. A challenge for oVRcome is that we use real world content for our VR videos as they are most effective, however creating such content is often difficult and expensive.

A typical example is VR content with airplane turbulence to help those with phobias of airplane flight. We would like many videos with a wide range of turbulence severity, however that is very difficult to film as such content cannot be created on demand. That is where GenAI video content creation is potentially very valuable. Models such as Veo3 show promise but are not a complete solution by themselves yet. oVRcome is investing in a technical pipeline that will create realistic clinically appropriate videos using the latest range of available AI tools.

Summary

The goal of this project is to determine what can be inferred from this signal. Firstly recreate existing results using deep learning and data science, then extend such results to the oVRcome dataset. For example, it is claimed that breathing can be detected by the PPG signal. We would like to verify this claim.

Many of these tools are not tested, and some just provide an academic paper with no code or AI model at all. oVRcome has a list of several potential tools that we want investigated and extended if appropriate. This project involves reproducing and extending such results and will likely include training or fine tuning AI models to achieve the desired results. Knowledge of CGI and video creation is desired but not essential.

Project Code	PHF
Type	DATA603 (Industry based internship)
Title	Antimicrobial resistance in the environment
Industry partner	New Zealand Institute for Public Health and Forensic Science (PHF)
Location	University / PHF Office (nearby)
Students	Could be either an individual or a group

Project Summary

Background

This summer internship is part of a larger, study of AMR in the environment. The student project will be mainly focused on analysing data that was collected in the field investigating the potential impact of climate change related factors (including temperature and salinity) on AMR in surface water during the year of collection.

Environmental data includes water pathogen data, water AMR gene data, and sediment measures. Main outputs will be reports that characterise the relationships between the environmental measures and AMR measures from field surveys (indicator pathogens, and AMR genes evident from metagenomics).

Data / resources provided

Overview

3 pilot sites and 5 study sites along the Heathcote River selected by PHF Science

~6 CCC sites

Quarterly sampling data from February sampling and August sampling.

Flood sampling data (awaiting rainfall of over 20mm for collection time).

Water only: Bacterial counts enterococci, E. coli, resistant E. coli; Metagenomic data for AMR genes and bacterial diversity.

Microbial abundances and AMR gene data

Data from whole water, sediment samples and positive colilerts (which enrich for things that grow in humans). n = 31 from different sites/types.

Bacterial counts enterococci, E. coli, resistant E. coli from water and sediment

Metagenomic data water and sediment, analysed for AMR genes and bacterial diversity

Logger data

~4000 daily temperature readings at hourly intervals for each site

~ 66000 flow readings every 5 minutes from data logger (at 1 site). Extra data can be obtained from <https://www.ecan.govt.nz/data/riverflow> is LAWA

232 rainfall daily readings (1/1/2025 to 20/08/2025)

- Chemical Analysis
- Sediment
- pH
- Heavy metal, trace level
As,Cd,Cr,Cu,Ni,Pb,Zn
- Total Recoverable Arsenic
- Total Recoverable Cadmium
- Total Recoverable Chromium
- Total Recoverable Copper
- Total Recoverable Lead
- Total Recoverable Nickel
- Total Recoverable Zinc

- Water:
- Sum of Anions
- Sum of Cations
- pH
- Total Alkalinity
- Bicarbonate
- Total Hardness
- Electrical Conductivity (EC)
- Approx Total Dissolved Salts
- Dissolved Calcium
- Dissolved Magnesium
- Dissolved Potassium
- Dissolved Sodium
- Chloride
- Total Nitrogen
- Nitrite-N
- Nitrate-N
- Nitrate-N + Nitrite-N
- Sulphate
- Non-Purgeable Organic Carbon (NPOC)
- Heavy metals, totals, trace
As,Cd,Cr,Cu,Ni,Pb,Zn
- Total Arsenic
- Total Cadmium
- Total Chromium
- Total Copper
- Total Lead
- Total Nickel
- Total Z

Skills required

- Data processing and matching
- Environmental measurement knowledge

Project Code	Plunket
Type	DATA603 (Industry based internship)
Title	Improving Client Relationships at Whānau Āwhina Plunket through Natural Language Processing and Large Language Models
Industry partner	Whānau Āwhina (Plunket)
Location	University of Canterbury
Students	Could be either an individual or a group

Project Summary

Background

Whānau Āwhina Plunket is a national charity dedicated to supporting whānau, ensuring their pēpi and tamariki have the best possible start in life. We see 80% of all newborns in Aotearoa New Zealand—including over half of all Māori pēpi.

Our core service is the Well Child Tamariki Ora programme, delivering essential health checks from 6 weeks to 5 years of age. These checks help identify developmental or health concerns early, setting children up for better outcomes. Another key service is PlunketLine: a free, 24/7 parenting and child health helpline, including video support for breastfeeding and sleep.

Data underpins everything we do. It guides our planning, drives continuous improvement, and ensures our services evolve to meet whānau needs. Equity is woven into all aspects of our work—but we know there’s more to learn, especially around supporting priority populations, including families living with disability, Māori, Pacific and Asian whānau

Summary

Over time, we’ve gathered rich insights through free text fields across our systems. While helpful in context, this data remains underutilised. This project aims to unlock its potential.

Using our Databricks platform, the intern will build a vector search index from our free text data. This will feed into a Retrieval-Augmented Generation (RAG) model, enabling frontline staff to interact with an LLM that surfaces relevant client context instantly.

The goal is to have an internal app which allows for better-informed conversations, stronger relationships with whānau, and more time for what matters. Strategically, our leadership team can also use it to learn what challenges our whānau face, that we may not have been known before.

The project is internship based, with a combination of working from home (WFH) and time spent at Christchurch’s Twigger St office. As Plunket is a national service with the Data & Insights team working across the country, the intern will travel to Wellington twice, once to meet the wider team, and again to present their work at the end of the project.

This project will suit someone who has a deep interest in AI Engineering/MLOps and strong programming skills.

Project Code	RuralCo1
Type	DATA605 (Industry based research project)
Title	Farmer Weather Analysis Tool
Industry partner	RuralCo
Location	University
Students	Group

Project Summary

Background

RuralCo is a values-led farming co-operative since 1963. They make farming life easier with the convenience of one card to use online or in store with one statement for consolidated spend & improved cashflow. Through their supplier network they offer competitive pricing with real value to meet the needs of farming businesses. They are experts in agriculture, providing a complete solution with personalised farming advice and the latest innovations, on farm, instore or online.

Problem statement

Build a tool that uses historic climatic information to help farmer forecasts what weather maybe this year and helps predict both what is best to grow and what farm inputs they may need for the following season. The tool would need to allow the farmer to input for their specific data as well to produce more specific results.

Project Data

RuralCo will supply all the data for the project. UC will supply a dedicated secure server for the project teams exclusive use, the project data will be uploaded to this server at the start of the project. The project data will be deleted at the end of the project before the server is decommissioned.

NDA

All students and project supervisors must sign an NDA before the start of the project.

Note

There will be regular on campus meetings with FarmCo employees to update them with project progress and have the opportunity to ask questions. Dates and times to be advised once project has started.

Project Code	RuralCo2
Type	DATA605 (Industry based research project)
Title	Customer Chance of Failure
Industry partner	RuralCo
Location	University
Students	Group

Project Summary

Background

RuralCo is a values-led farming co-operative since 1963. They make farming life easier with the convenience of one card to use online or in store with one statement for consolidated spend & improved cashflow. Through their supplier network they offer competitive pricing with real value to meet the needs of farming businesses. They are experts in agriculture, providing a complete solution with personalised farming advice and the latest innovations, on farm, instore or online.

Problem statement

Need to tool that consistently evaluates all customers and gives the risk of failure, along with a potential failure timeframe. The tool should consider:

- Payment History
- Spend pattern, how are they using their credit limit – for business inputs or for personal needs ie food
- External credit scores
- Age / Location / Length of time in business
- Industry
- Links to any external factors ie weather events / dairy payouts
- Consider chance of failure for first time revolver v long term revolver.

FISERV has tools to support this.

Project Data

RuralCo will supply all the data for the project. UC will supply a dedicated secure server for the project teams exclusive use, the project data will be uploaded to this server at the start of the project. The project data will be deleted at the end of the project before the server is decommissioned.

NDA

All students and project supervisors must sign an NDA before the start of the project.

Note

There will be regular on campus meetings with FarmCo employees to update them with project progress and have the opportunity to ask questions. Dates and times to be advised once project has started.

Project Code	RuralCo3
Type	DATA605 (Industry based research project)
Title	Revenue Opportunity for retention of Discounts
Industry partner	RuralCo
Location	University
Students	Group

Project Summary

Background

RuralCo is a values-led farming co-operative since 1963. They make farming life easier with the convenience of one card to use online or in store with one statement for consolidated spend & improved cashflow. Through their supplier network they offer competitive pricing with real value to meet the needs of farming businesses. They are experts in agriculture, providing a complete solution with personalised farming advice and the latest innovations, on farm, instore or online.

Problem statement

Previously Ruralco retained discounts earned by members when payment wasn't made. This practice stopped some 20 years ago, the business is interested in what the Revenue opportunity would look like if we brought this practice back. We would need to review the records of non payers for last 18 months, analyse the invoices and determine discounts received. As some merchants put the discount on up front we may have to reverse engineer some of these discount items.

Project Data

RuralCo will supply all the data for the project. UC will supply a dedicated secure server for the project teams exclusive use, the project data will be uploaded to this server at the start of the project. The project data will be deleted at the end of the project before the server is decommissioned.

NDA

All students and project supervisors must sign an NDA before the start of the project.

There will be regular on campus meetings with FarmCo employees to update them with project progress and have the opportunity to ask questions. Dates and times to be advised once project has started.

Project Code	Scion
Type	DATA603 (Industry based internship)
Title	Development and Application of a Geospatial-Temporal Data Analysis Interface for the Wildfire Occurrence Database
Industry partner	Scion
Location	University of Canterbury
Students	Individual

Project Summary

Background

At Scion, we believe the power of forests can transform the way we live. Through science, research, and innovation, we work across sectors with industry, government, and iwi to help lead New Zealand's transition to a circular bioeconomy and achieve a net-zero 2050 and beyond. We are impact-driven organisation focused on delivering positive environmental, social and economic outcomes for New Zealand. As a Crown Research Institute (CRI) for forestry in New Zealand, our purpose is mandated by the Crown: "To drive innovation and growth from New Zealand's forestry, industrial biotechnology and advanced manufacturing sectors to create economic value and contribute to beneficial environmental and social outcomes for New Zealand." For more than 75 years, we have led research that creates resilient planted forests, productive timber industries and sustainable product solutions.

Scion staff will provide regular virtual meetings and guidance throughout the project. Due to an ongoing office relocation, in-person work is not currently possible so the position will be remotely based.

Problem statement

This project aims to address this gap by developing a user-friendly interface that enables exploration, visualization, and analysis of the database, incorporating its geospatial and temporal nature at multiple scales. Once the interface is created, it can be applied to answer research questions, developed together with the student.

Scope of work

At Scion we have created the Wildfire Occurrence Database – a large scale database of historical fire records and environmental conditions, which will soon be available to the public. While this database can provide many insights into wildfire occurrence in New Zealand, it's size can make it difficult to access the data without programming knowledge.

Deliverables

User-friendly interface

Data / resources provided

Wildfire Occurrence Database

Skills required

Scion would prefer for the student to have experience working with geospatial data. Scion

Project Code	Search
Type	DATA601 (Academic research project)
Title	Semantic Search Engine for Academic Research Papers
Department	University of Canterbury (Maths and Stats)
Location	University of Canterbury
Students	Group

Project Summary

Overview

The goal is to build a semantic search engine that allows users to query a collection of academic research papers and retrieve relevant results based on the meaning of the query, rather than keyword matching. This project leverages Pinecone for vector similarity search, natural language processing (NLP) for text embedding, and a user-friendly interface for interaction. It demonstrates proficiency in data pre-processing, machine learning and vector databases.

Why this Project?

- **Relevance:** Semantic search is a cutting-edge application in data science, used in industries like academia, e-commerce, and content discovery.
- **Pinecone Integration:** Pinecone's sample apps (e.g., semantic search for text) provide a foundation for implementing vector-based search, making it accessible for students.
- **Team Size:** The project has distinct components (data collection, pre-processing, embedding, Pinecone integration, and a UI) that can be divided among 3-4 students.
- **Impact:** The tool can assist students and researchers in finding relevant papers, addressing a practical need.
- **Feasibility:** It uses open-source datasets and pre-trained models, reducing the need for extensive computational resources.

Project Components and Workflow

The project can be broken down into manageable tasks, each suitable for a team member to lead while collaborating on integration.

1. **Data Collection and Preprocessing (Assigned to Student 1)**
 - **Task:** Gather a dataset of academic research papers (e.g., abstracts or full texts) from open-source repositories like arXiv, PubMed, or Semantic Scholar.
 - **Dataset Example:** Use the arXiv API to download metadata and abstracts for 10,000–50,000 papers in a specific domain (e.g., computer science or biology).
 - **Preprocessing:**
 - Clean text data (remove special characters, normalize case).
 - Extract relevant fields (title, abstract, keywords).
 - Store metadata in a structured format (e.g., JSON or CSV).
 - **Tools:** Python, Pandas, NLTK/spaCy for text cleaning, arXiv API.
 - **Deliverable:** A clean dataset ready for embedding.
2. **Text Embedding with NLP Models (Assigned to Student 2)**
 - **Task:** Convert paper abstracts into dense vector representations using a pre-trained NLP model.

- Model Choice: Use a model like Sentence-BERT (SBERT) or Hugging Face's `all-MiniLM-L6-v2` for generating embeddings. These models are lightweight and effective for semantic similarity tasks, as seen in Pinecone's semantic search sample app.
 - Process:
 - Load the pre-processed dataset.
 - Generate embeddings for each abstract (e.g., 384-dimensional vectors for MiniLM).
 - Ensure embeddings capture semantic meaning (e.g., similar papers have closer vectors).
 - Tools: Python, Hugging Face Transformers, NumPy.
 - Deliverable: A set of vector embeddings for all papers.
3. Pinecone Vector Database Integration (Assigned to Student 3)
- Task: Store embeddings in Pinecone and implement a query system for semantic search.
 - Pinecone Setup:
 - Create a Pinecone index with the appropriate dimensionality (e.g., 384 for MiniLM).
 - Upsert embeddings with metadata (e.g., paper ID, title) into the Pinecone index, following Pinecone's sample app structure for text search.
 - Implement a query function that:
 - Converts user queries into embeddings using the same NLP model.
 - Queries the Pinecone index to retrieve the top-k most similar papers.
 - Tools: Pinecone Python SDK, Python.
 - Deliverable: A functional semantic search backend using Pinecone.
4. UI Development and Visualization (Assigned to Student 4 or Shared)
- Task: Build a simple web interface for users to input queries and view search results.
 - Interface:
 - Create a web app using Streamlit or Flask or Shiny where users can enter a natural language query (e.g., "machine learning for healthcare").
 - Display results with paper titles, abstracts, and similarity scores.
 - Optionally, include visualizations (e.g., a 2D t-SNE plot of paper embeddings to show clustering).
 - Tools: Shiny/Streamlit/Flask, Plotly/Seaborn for visualization, HTML/CSS for basic styling.
 - Deliverable: A user-friendly interface for interacting with the search engine.

Implementation Steps

The following is a suggested approach to implementation:

1. Data Collection: Obtain customer datasets from Kaggle, such as:
 - <https://www.kaggle.com/datasets/vjchoudhary7/customer-segmentation-tutorial-in-python>
 - <https://www.kaggle.com/datasets/kandij/mall-customers>
 - <https://www.kaggle.com/datasets/imakash3011/customer-personality-analysis>
 - <https://www.kaggle.com/code/halimedogan/customer-segmentation-using-rfm/input>
2. Pre-processing: Clean data (handle missing values, outliers), encode categorical variables, normalize numerical features, and extract text-based features (e.g., sentiment from reviews) using an LLM or more traditional manual techniques.
3. Clustering:
 - Apply clustering algorithms (e.g., K-means, hierarchical clustering) to segment customers.
 - Use dimensionality reduction (e.g., ICA, PCA, t-SNE) for visualization and feature optimization.
4. Interpretation: Prompt the LLM to generate human-readable descriptions of each cluster based on feature distributions and customer profiles.
5. Recommendation: Use the LLM to create marketing strategies for each segment (e.g., discounts for price-sensitive customers, premium offers for high-spenders).

6. Evaluation: Assess clustering quality using metrics like silhouette score, Davies-Bouldin index, and business relevance (e.g., segment distinctiveness). Validate LLM-generated outputs via human review.
7. Visualization and Deployment: Build an interactive interface (e.g., Shiny, Streamlit, Flask) to display segment profiles, visualizations (e.g., cluster scatter plots), and marketing recommendations.

Team Roles and Collaboration

Here's a proposed role breakdown, which may vary based on the team's skills and strengths.

- Student 1 (Data Engineer): Leads data collection and preprocessing, ensures data quality.
- Student 2 (NLP Specialist): Handles embedding generation, validates semantic quality.
- Student 3 (Database Engineer): Manages Pinecone integration, optimizes search performance.
- Student 4 (UI Developer): Develops the UI, integrates backend with frontend.
- Collaboration: All students collaborate on integrating components (e.g., passing embeddings to Pinecone, connecting search results to the UI) and testing the end-to-end system.

Timeline

Here's a proposed project timeline.

- Weeks 1-2: Project planning, dataset selection, and environment setup.
- Weeks 3-4: Data collection and pre-processing.
- Weeks 5-6: Embedding generation and Pinecone index setup.
- Weeks 7-8: Query system implementation and frontend development.
- Weeks 9-10: System integration, testing, and debugging.
- Weeks 11-12: Final refinements, documentation, and presentation preparation.

Technical Requirements

These are suggested tools and technologies; feel free to use alternatives that you find more suitable or are more comfortable with.

- Programming: Python for data processing, NLP, and Pinecone integration; HTML/CSS/JavaScript for frontend (if using Flask).
- Libraries: Pandas, NumPy, Hugging Face Transformers, Pinecone SDK, Streamlit/Flask, Shiny, Plotly/Seaborn.
- Hardware: A standard laptop for preprocessing and embedding; Pinecone's cloud service handles vector storage and search, reducing local compute needs.
- Pinecone Account: Free tier for prototyping, with potential upgrade to starter plan for larger datasets (check Pinecone's pricing).

Learning Outcomes

- Technical Skills: Data preprocessing, NLP, vector databases, web development.
- Soft Skills: Teamwork, project management, communication.
- Portfolio Impact: A tangible, industry-relevant project showcasing end-to-end data science skills.

Resources

Here are a list of resources that you may find as useful starting points for your research.

- Open-Source Datasets: arXiv, PubMed, Semantic Scholar for academic papers.
- Pinecone Documentation: Sample apps (e.g., semantic text search) for implementation guidance.
- Hugging Face: Pre-trained NLP models.
- Streamlit/Flask Tutorials: For web app development.

Deliverables

For detailed requirements of the project report and presentation, please consult LEARN.

- A working semantic search engine hosted locally or on a cloud platform (e.g., Heroku for Flask apps).
- A final report detailing the data, methodology, challenges, and results.
- A presentation showcasing the tool, demoing a few queries, and discussing technical insights.

Potential Challenges and Mitigations

- **Data Quality:** Ensure the dataset is clean and relevant; filter out incomplete records during pre-processing.
- **Embedding Quality:** Test embeddings with sample queries to verify semantic accuracy; adjust model if needed.
- **Pinecone Limits:** Monitor usage to stay within the free tier or request academic credits from Pinecone.
- **Team Coordination:** Use tools like GitHub for code sharing and Trello/Slack for task management.
- **Time Management:** Break tasks into milestones and assign clear responsibilities.

Scalability and Extensions

If time permits, the team could:

- Expand the dataset to include multiple domains (e.g., biology and physics).
- Add features like filtering by publication year or journal.
- Implement a recommendation system for related papers based on search history.
- Deploy the app on a public cloud for broader access.

Documentation and Iterative Improvement

Students should maintain detailed documentation of their methodology, challenges, and iterative refinements to demonstrate the development process. Regular updates to the project report and code comments will enhance transparency and showcase learning outcomes.

Project Code	TANZ1
Type	DATA603 (Industry based internship)
Title	AI, Graph RAG and Knowledge Graphs
Industry partner	Trust Alliance New Zealand (TANZ)
Location	Christchurch
Students	Individual

Project Summary

Background

Trust Alliance NZ Inc. (TANZ) was established in August 2021 to enable trusted data sharing. TANZ aims to support New Zealand industry by reducing costs and friction in the market and ensuring value to all participants; and providing the primary and related industries with a digital identity framework enabling data sovereignty for farmers.

Trust Alliance New Zealand (TANZ) has developed the Digital Farm Wallet (DFW), a modular and standards-aligned service offering that enables the verified and secure sharing of farm data to support regional consenting and compliance processes. At the heart of the service is a digital farm register that captures key attributes such as ownership, land title, location, and digital proofs including water usage and nitrate loss risk. This enables verified farmers to seamlessly share their environmental performance data with Environment Canterbury (ECAN), streamlining consent workflows and reducing administrative burden.

A test farm registry has been created in PostgreSQL. The plan is to populate the registry with publicly available data. A series of tools need to be developed to allow this to occur. These include knowledge graphs, Graph RAG tools and MCP servers.

A knowledge graph is an essential tool for providing context for AI. Neo4J provides tools for creating knowledge graphs and utilising with AI tools. MCP servers provide a mechanism to populate the knowledge graph with structured and unstructured data.

The premise is to allow the farmers to claim and validate the data objects that are publicly available. To then modify, store and transmit using tools such as digital wallets. Further, TANZ plans to generate, using AI, regulatory objects such fresh water farm plans; and consent applications for farm activities such as irrigation.

Fresh Water Farm plans are a tool designed to stop further decline in freshwater quality, make water quality improvements within five years, and begin to reverse past damage to our waterways. They will identify practical on-farm actions to help improve local waterways. These will depend on the farm's location, catchment and local environment, and will provide a more flexible alternative to consents and one-size-fits-all approaches.

In time, Freshwater Farm Plans are expected to become the central tool for farmers to manage all freshwater requirements.

Under current regulations, all farms with 20 hectares or more in arable, pastoral or combined use, or five hectares or more in horticultural use will eventually require a FWFP. All FWFPs will need to be certified and audited.

Farms with specific irrigation requirements will still need to prepare and apply for an irrigation consent. There are currently 28,000 live consents issued by Environment Canterbury under the current Resource Management Act (RMA).

Problem statement

By 2030, Environment Canterbury is expected to receive up to 1,500 farming consent renewals per year, creating pressure on its current staffing and systems. The existing process for renewing consents is not efficient or scalable, and there is no cost-effective way to collect on-farm environmental data for regulatory use. TANZ seeks to reduce the compliance burden on farmers and the administrative burden on regulators by improving how farm environmental data is collected and used during the consent process.

Scope of work

Using Neo4J MCP & Claude (or similar tools) to create a labelled property graph (knowledge graph) of the farm identity and the elements required for a farm consent for irrigation.

Using Neo4J & Claude (or similar tools) to retrieve and create a catalogue of farm consent conditions for the target farms that could then be used to generate a consent renewal application and a Fresh Water Farm Plan.

Deliverables

The student will, using the farm identity entity relationship diagram, consent condition information and Neo4j's MCP with Claude Desktop (or similar tools) retrieve farm consent condition data from unstructured/structured data provided to generate a consent renewal & Fresh Water Farm Plan FWFP.

Data/Resources Available

- Data sets provided
- Basic Neo4J set up
- Test environment
- Physical location – Backstory worklab
- Technical mentors and support

Skills required

- Basic understanding of Knowledge graphs and deployment of online tools.

References

environment.govt.nz/acts-and-regulations/freshwater-implementation-guidance/freshwater-farm-plans/
ecan.govt.nz/your-region/farmers-hub/farming-plans-and-consenting/freshwater-farm-plans

Project Code	TANZ2
Type	DATA603 (Industry based internship)
Title	AI & Knowledge Graphs for Water, Carbon and Biodiversity
Industry partner	Trust Alliance New Zealand (TANZ)
Location	Christchurch
Students	Individual

Project Summary

Background

Trust Alliance NZ Inc. (TANZ) was established in August 2021 to enable trusted data sharing. TANZ aims to support New Zealand industry by reducing costs and friction in the market and ensuring value to all participants; and providing the primary and related industries with a digital identity framework enabling data sovereignty for farmers.

Trust Alliance New Zealand (TANZ) has developed the Digital Farm Wallet (DFW), a modular and standards-aligned service offering that enables the verified and secure sharing of farm data to support regional consenting and compliance processes. At the heart of the service is a digital farm register that captures key attributes such as ownership, land title, location, and digital proofs including water usage and nitrate loss risk. This enables verified farmers to seamlessly share their environmental performance data with Environment Canterbury (ECAN), streamlining consent workflows and reducing administrative burden.

A test farm registry has been created in PostgreSQL. The plan is to populate the registry with publicly available data. A series of tools need to be developed to allow this to occur. These include knowledge graphs, Graph RAG tools and MCP servers.

A knowledge graph is an essential tool for providing context for AI. Neo4J provides tools for creating knowledge graphs and utilising with AI tools. MCP servers provide a mechanism to populate the knowledge graph with structured and unstructured data.

The premise is to allow the farmers to claim and validate the data objects that are publicly available. To then modify, store and transmit using tools such as digital wallets. Further, TANZ plans to generate, using AI, regulatory objects such as fresh water farm plans; and consent applications for farm activities such as irrigation.

Fresh Water Farm plans are a tool designed to stop further decline in freshwater quality, make water quality improvements within five years, and begin to reverse past damage to our waterways. They will identify practical on-farm actions to help improve local waterways. These will depend on the farm's location, catchment and local environment, and will provide a more flexible alternative to consents and one-size-fits-all approaches.

In time, Freshwater Farm Plans are expected to become the central tool for farmers to manage all freshwater requirements.

Under current regulations, all farms with 20 hectares or more in arable, pastoral or combined use, or five hectares or more in horticultural use will eventually require a FWFP. All FWFPs will need to be certified and audited.

Farms with specific irrigation requirements will still need to prepare and apply for an irrigation consent. There are currently 28,000 live consents issued by Environment Canterbury under the current Resource Management Act (RMA).

Problem statement

By 2030, Environment Canterbury is expected to receive up to 1,500 farming consent renewals per year, creating pressure on its current staffing and systems. The existing process for renewing consents is not efficient or scalable, and there is no cost-effective way to collect on-farm environmental data for regulatory use. TANZ seeks to reduce the compliance burden on farmers and the administrative burden on regulators by improving how farm environmental data is collected and used during the consent process.

Scope of work

The student will use Neo4J MCP & Claude (or similar tools) to create a labelled property graph (knowledge graph) of the farm identity and the elements required for a farm consent for irrigation, and identify how to expand the knowledge graph to include other farm data attributes ie carbon, biodiversity and farm quality assurance models.

Deliverables

Using the farm identity entity relationship diagram, consent condition information and Neo4j's MCP with Claude Desktop (or similar tools) generate knowledge graphs from unstructured/structured data provided.

Demonstrate the knowledge graph working and how the entity relationships can be expanded to include irrigation consent conditions.

Populate the knowledge graph with test data and show how data could be added from the publicly available ECAN consent dataset.

Data/Resources Available

- Data sets provided
- Basic Neo4J set up
- Test environment
- Physical location – Backstory worklab
- Technical mentors and support

Skills required

- Basic understanding of Knowledge graphs and deployment of online tools.

References

environment.govt.nz/acts-and-regulations/freshwater-implementation-guidance/freshwater-farm-plans/
ecan.govt.nz/your-region/farmers-hub/farming-plans-and-consenting/freshwater-farm-plans

Project Code	TANZ3
Type	DATA603 (Industry based internship)
Title	Neo4J MCP Servers & NZ Government Data
Industry partner	Trust Alliance New Zealand (TANZ)
Location	Christchurch
Students	Individual

Project Summary

Background

Trust Alliance NZ Inc. (TANZ) was established in August 2021 to enable trusted data sharing. TANZ aims to support New Zealand industry by reducing costs and friction in the market and ensuring value to all participants; and providing the primary and related industries with a digital identity framework enabling data sovereignty for farmers.

Trust Alliance New Zealand (TANZ) has developed the Digital Farm Wallet (DFW), a modular and standards-aligned service offering that enables the verified and secure sharing of farm data to support regional consenting and compliance processes. At the heart of the service is a digital farm register that captures key attributes such as ownership, land title, location, and digital proofs including water usage and nitrate loss risk. This enables verified farmers to seamlessly share their environmental performance data with Environment Canterbury (ECAN), streamlining consent workflows and reducing administrative burden.

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In time, Freshwater Farm Plans are expected to become the central tool for farmers to manage all freshwater requirements.

Under current regulations, all farms with 20 hectares or more in arable, pastoral or combined use, or five hectares or more in horticultural use will eventually require a FWFP. All FWFPs will need to be certified and audited.

Farms with specific irrigation requirements will still need to prepare and apply for an irrigation consent. There are currently 28,000 live consents issued by Environment Canterbury under the current Resource Management Act (RMA).

Problem statement

By 2030, Environment Canterbury is expected to receive up to 1,500 farming consent renewals per year, creating pressure on its current staffing and systems. The existing process for renewing consents is not efficient or scalable, and there is no cost-effective way to collect on-farm environmental data for regulatory use. TANZ seeks to reduce the compliance burden on farmers and the administrative burden on regulators by improving how farm environmental data is collected and used during the consent process.

Scope of work

Using Neo4J MCP & Claude (or similar tools) to create a labelled property graph (knowledge graph) of the farm identity and the elements required for a farm consent for irrigation.

Using Neo4J MCP server link via the Government API Gateway to relevant data sets and populate the knowledge graph to validate the existing data set.

Automate the data retrieval to show how the process could be scaled.

Demonstrate how publicly available data ie NZ Business Numbers (NZBN) can be used to provide digital trust anchors.

Deliverables

The student will, using the farm identity entity relationship diagram, consent condition information and Neo4j's MCP with Claude Desktop (or similar tools) generate knowledge graphs from unstructured/structured data provided.

Populate the knowledge graph with test data and show how data could be added from the publicly available govt dataset.

Data / resources provided

- Data sets provided
- Basic Neo4J set up
- Test environment
- Physical location – Backstory worklab
- Technical mentors and support

Skills required

- Basic understanding of Knowledge graphs and deployment of online tools.

References

environment.govt.nz/acts-and-regulations/freshwater-implementation-guidance/freshwater-farm-plans/
ecan.govt.nz/your-region/farmers-hub/farming-plans-and-consenting/freshwater-farm-plans

Project Code	TANZ4
Type	DATA603 (Industry based internship)
Title	AI & User Interfaces to Knowledge Graphs
Industry partner	Trust Alliance New Zealand (TANZ)
Location	Christchurch
Students	Individual

Project Summary

Background

Trust Alliance NZ Inc. (TANZ) was established in August 2021 to enable trusted data sharing. TANZ aims to support New Zealand industry by reducing costs and friction in the market and ensuring value to all participants; and providing the primary and related industries with a digital identity framework enabling data sovereignty for farmers.

Trust Alliance New Zealand (TANZ) has developed the Digital Farm Wallet (DFW), a modular and standards-aligned service offering that enables the verified and secure sharing of farm data to support regional consenting and compliance processes. At the heart of the service is a digital farm register that captures key attributes such as ownership, land title, location, and digital proofs including water usage and nitrate loss risk. This enables verified farmers to seamlessly share their environmental performance data with Environment Canterbury (ECAN), streamlining consent workflows and reducing administrative burden.

A test farm registry has been created in PostgreSQL. The plan is to populate the registry with publicly available data. A series of tools need to be developed to allow this to occur. These include knowledge graphs, Graph RAG tools and MCP servers.

A knowledge graph is an essential tool for providing context for AI. Neo4J provides tools for creating knowledge graphs and utilising with AI tools. MCP servers provide a mechanism to populate the knowledge graph with structured and unstructured data.

The premise is to allow the farmers to claim and validate the data objects that are publicly available. To then modify, store and transmit using tools such as digital wallets. Further, TANZ plans to generate, using AI, regulatory objects such fresh water farm plans; and consent applications for farm activities such as irrigation.

Fresh Water Farm plans are a tool designed to stop further decline in freshwater quality, make water quality improvements within five years, and begin to reverse past damage to our waterways. They will identify practical on-farm actions to help improve local waterways. These will depend on the farm's location, catchment and local environment, and will provide a more flexible alternative to consents and one-size-fits-all approaches.

In time, Freshwater Farm Plans are expected to become the central tool for farmers to manage all freshwater requirements.

Under current regulations, all farms with 20 hectares or more in arable, pastoral or combined use, or five hectares or more in horticultural use will eventually require a FWFP. All FWFPs will need to be certified and audited.

Farms with specific irrigation requirements will still need to prepare and apply for an irrigation consent. There are currently 28,000 live consents issued by Environment Canterbury under the current Resource Management Act (RMA).

Problem statement

By 2030, Environment Canterbury is expected to receive up to 1,500 farming consent renewals per year, creating pressure on its current staffing and systems. The existing process for renewing consents is not efficient or scalable, and there is no cost-effective way to collect on-farm environmental data for regulatory use. TANZ seeks to reduce the compliance burden on farmers and the administrative burden on regulators by improving how farm environmental data is collected and used during the consent process.

Scope of work

Using Neo4J MCP & Claude (or similar tools) to create a labelled property graph (knowledge graph) of the farm identity and the elements required for a farm consent for irrigation.

Create a POC of a user interface to knowledge graph – ie online portal that a farmer can claim objects related to them ie consents etc.

Show how the farmer can then modify the elements in the knowledge graph and provide as evidence to another party.

Deliverables

The student will, using the farm identity entity relationship diagram, consent condition information and Neo4j's MCP with Claude Desktop (or similar tools) generate knowledge graphs from unstructured/structured data provided.

Let a user claim and modify the datasets to provide evidence to a third party.

Data / resources provided

- Data sets provided
- Basic Neo4J set up
- Test environment
- Physical location – Backstory worklab
- Technical mentors and support

Skills required

- Basic understanding of Knowledge graphs and deployment of online tools.

References

environment.govt.nz/acts-and-regulations/freshwater-implementation-guidance/freshwater-farm-plans/
ecan.govt.nz/your-region/farmers-hub/farming-plans-and-consenting/freshwater-farm-plans

Project Code	UCT
Type	DATA601 (Academic research project)
Title	Modelling access to UC by different transport modes
Department	School of Earth and Environment
Location	University of Canterbury
Students	Could be an individual or a group

Project Summary

Background

According to a 2024 University of Canterbury survey, 35% of students and staff travelled to campus by car, 21% cycled, 20% walked, and 15% took the bus. Over recent years, the University has made significant efforts to encourage sustainable modes of transport—such as walking, cycling, and public transport—leading to a gradual decline in car use. However, with around 20,000 people attending UC each week and approximately 35% still travelling by car, on-campus parking (3,500 spaces) is frequently at capacity.

Problem statement

Despite ongoing initiatives, a substantial number of students and staff continue to rely on private vehicles, partly due to limited awareness of viable alternative transport options. Many people are unaware of nearby cycle routes, bus routes, and realistic travel times by alternative modes. The challenge is to better understand and communicate how accessible the University is by different transport modes and to support informed decision-making about travel and accommodation choices.

Scope of work / expected methodology

The project will model access to the University of Canterbury by different transport modes, including car, bus, bicycle, and walking.

Key tasks include:

- Integrating and analysing geospatial data such as bus routes, timetables, cycleways, road networks.
- Incorporating additional contextual data such as speed limits, one-way streets, cycle ‘short cuts’.
- Building a data model or accessibility index to compare travel options by time and distance.
- Developing a prototype map-based dashboard / visualization tool that students and staff can use to identify their options for travelling to the University and to locate places to live based on access.

Deliverables

- A geospatial dataset and/or model describing access to UC by different transport modes.
- A map-based dashboard visualising travel options and accessibility from different areas.
- Documentation explaining data sources, methodology, and potential applications for transport planning or accommodation decisions.

Data / resources provided

- Existing datasets on bus routes, timetables, and cycleways.
- Access to internal UC geospatial or transport datasets where available.
- Guidance from University staff involved in transport planning and sustainability.
- Support in identifying and collating additional public datasets (e.g. road speed limits, one-way streets).

Project Code	Waitaki
Type	DATA603 (Industry based internship)
Title	Waitaki Biosciences data migration, verification, and modeling
Industry partner	Waitaki Biosciences
Location	University of Canterbury
Students	Could be either an individual or a group

Project Summary

Background

Waitaki Biosciences is the manufacturing and ingredient-development arm of PharmaZen Ltd, headquartered in Christchurch, New Zealand. The company specializes in natural nutritional ingredients for nutraceuticals, dietary supplements, personal care and animal health markets. Their products derive from New Zealand’s meat, marine, botanical and agricultural sectors, using techniques such as freeze-drying, proprietary digestion or extraction, oil extraction, and formulation into finished goods (capsules, gummies, pet treats).

Waitaki operates with strict emphasis on quality, traceability, food safety, and regulatory compliance. It holds multiple licenses and certifications (MPI / RMP, ISO 9001, food safety standards, export compliance) to serve global markets in the US, Europe, Asia, Africa and Oceania. In recent years, the company has invested heavily in capacity expansion (additional freeze dryers, extraction facilities) to meet increasing demand and new regulatory standards (including GMP / pharmaceutical-grade manufacturing).

Given this growth and complexity, robust management of product data—covering historical records, quality thresholds, Certificates of Analysis (CoAs), and analytical / trend modelling—is critical for supporting launches, compliance, traceability, and risk monitoring across multiple product lines and markets.

Problem statement

Waitaki is in the process of **deploying a new software system** for managing product data, which is central to product release, regulatory compliance, and operational visibility. The challenge is multi-faceted:

1. **Historical Data Migration:** 3–5 years’ worth of legacy product data must be imported into the new system reliably and with minimal loss or corruption.
2. **Data Verification / Integrity:** The imported data must be audited and corrected where needed, ensuring that thresholds, analytical values, CoA data, and metadata are accurate and consistent.
3. **Data Modelling & Analytics:** Beyond data storage, the system should support trend detection, risk identification, and opportunities for growth by linking product performance, quality variability, raw material sources, and market feedback.

Scope of work

Work Area	Tasks / Activities
Data collection & ingestion	Inventory all existing product datasets (raw materials, production batches, quality tests, CoAs, product metadata). Extract them (from legacy systems, spreadsheets, lab systems).
Data cleaning & verification	Profile data quality (missing values, outliers, inconsistencies). Cross-validate thresholds and CoA fields with lab / QA records. Flag and resolve discrepancies with subject matter experts.
Validation & documentation	Document data lineage, transformation logic, validation rules. Develop test plans. Train users.

Deliverables

- Cleansed, validated, migrated historical product dataset (3–5 years)
- Code / scripts used for data transformation, validation, and modelling, with version control.

Skills

- Strong data engineering / ETL skills: data migration, transformation, cleaning, versioning.
- Experience in relational database design or other structured data modelling.
- Experience in statistical analysis, trend detection, anomaly detection or data modelling techniques.
- Proficiency in a programming or scripting language (Python, R, SQL, etc.).

Project Code	Wastewater1
Type	DATA601 (Academic research project)
Title	Covid-19 Wastewater Site Analysis
Department	School of Mathematics and Statistics
Academic supervisor	Leighton Watson and Mike Plank (leighton.watson@canterbury.ac.nz)
Location	University of Canterbury
Students	Suitable for either individual or group

Project Summary

Background

The New Zealand Institute for Public Health and Forensic Science Ltd (PHF Science) tests wastewater for the presence of SARS-CoV-2 (the virus that causes COVID-19). From 2020-2022, this aided Aotearoa's ability to identify outbreaks, and is now used to understand national trends and to monitor variants. To inform on trends, the raw wastewater data (genome copies per litre of wastewater) is converted to a viral load of genome copies per person per day (GC/person/day).

Problem statement

It is expensive and time consuming to sample wastewater data from sites all over the country. How can we pick the fewest sites necessary and coarsest sampling interval but still obtain useful information from the wastewater data?

Expected methodology

This project will involve analysing the SARS-CoV-2 wastewater data (which is available here: https://github.com/ESR-NZ/covid_in_wastewater) and building a time series forecasting model to forecast cases from the wastewater data. We could do this nationally or split into smaller regions.

Then we will down sample the number of wastewater sampling sites and the frequency of sampling to determine how the utility or predictive power of the wastewater data changes with the number of sites.

Deliverables

This project will address the following questions:

- How many sites are needed for wastewater data to be useful? What is the minimum possible number of sites and where are they located?
- What is the minimum sampling period needed for the data? Do we need to sample daily? Or can we sample once a week or even less frequently and still obtain useful information?

Data / resources provided

SARS-CoV-2 data (genome copies, GC per litre) provided by PHF Science. Number of reported cases. Population data. Catchment maps.

Skills

We have existing code for analysis wastewater and reported case data at the same time written in Julia (see References for more details on this). Therefore, the ability to program in Julia could be an advantage.

References

Watson, L.M., Plank, M.J., Armstrong, B.A. *et al.* Jointly estimating epidemiological dynamics of Covid-19 from case and wastewater data in Aotearoa New Zealand. *Commun Med* **4**, 143 (2024).
<https://doi.org/10.1038/s43856-024-00570-3>

Other information

This project will be run as DATA601 but will have opportunities for engagement with PHF Science staff, who are based next to UC campus.

Project Code	Wastewater2
Type	DATA601 (Academic research project)
Title	Covid-19 Wastewater Normalisation
Department	School of Mathematics and Statistics
Academic supervisor	Leighton Watson and Mike Plank (leighton.watson@canterbury.ac.nz)
Location	University of Canterbury
Students	Suitable for either individual or group

Project Summary

Background

The New Zealand Institute for Public Health and Forensic Science Ltd (PHF Science) tests wastewater for the presence of SARS-CoV-2 (the virus that causes COVID-19). From 2020-2022, this aided Aotearoa's ability to identify outbreaks, and is now used to understand national trends and to monitor variants. To inform on trends, the raw wastewater data (genome copies per litre of wastewater) is converted to a viral load of genome copies per person per day (GC/person/day). This conversion considers the flow of wastewater entering the treatment plant and the population in the wastewater catchment. At the site level, GC/person/day is the average value of all samples collected within that week. This approach allows for aggregation at site and national levels. The data is made available on a public dashboard.

While per capita normalisation has been shown to be useful, there are other approaches that can be used for normalisation. One is Pepper Mild Mottle Virus (PMMoV). PMMoV is a plant virus ingested through foods infected with it and is shed in high numbers in faeces. A number of labs utilise the PMMoV data to normalise the SARS-CoV-2 data. While PHF Science does not use PMMoV for this application, it is tested for and used as a control. An alternative is no normalization with or without statistical smoothing, combination of capita with PMMoV normalization, or other approaches.

Problem statement

Wastewater data is an indicator of infections in the community, and useful when other surveillance is lacking. However, wastewater data can be highly variable, and the levels of SARS-CoV-2, or other targets, can be affected by external factors too such as rain which may dilute the samples. While accurate flow data is collected by PHF Science, population estimates may be unreliable, and change over time (seasons, population movements, lack of updated census information) so the per capita normalisation may not be accurate. As the amount of PMMoV is diet related (peppers and pepper products), this may also be unreliable.

The goal of this project is to compare and contrast different normalisation methodologies for SARS-CoV-2 wastewater data.

Expected methodology

This project will involve analysing the SARS-CoV-2 wastewater data (which is available here: https://github.com/ESR-NZ/covid_in_wastewater) and applying different normalisation methodologies. Wastewater data will then be compared with reported cases and hospitalisation data to determine which normalisation approach is preferred.

Deliverables

- How different normalization approaches influence the output
- Recommendations to PHF Science on the preferred approach
- Sharing the data and analysis codes with PHF Science (so making sure your work is well documented and able to be passed over at the end of the project)

Data / resources provided

SARS-CoV-2 data (genome copies, GC per litre) and PMMoV data (GC/L) provided by PHF Science. Number of reported cases. Flow data and population data (population data to be updated?). Catchment maps.

References

Watson, L.M., Plank, M.J., Armstrong, B.A. *et al.* Jointly estimating epidemiological dynamics of Covid-19 from case and wastewater data in Aotearoa New Zealand. *Commun Med* **4**, 143 (2024).

<https://doi.org/10.1038/s43856-024-00570-3>

Other information

This project will be run as DATA601 but will have opportunities for engagement with PHF Science staff, who are based next to UC campus.

Project Code	WorldView1
Type	DATA601 (Academic research project)
Title	Develop the WorldView Shiny App to integrate and explore cross-national datasets with the World Values Survey
Department	School of Psychology, Speech and Hearing
Academic supervisor	Joseph Watts (joseph.watts@canterbury.ac.nz)
Location	University of Canterbury
Students	Individual

Project Summary

Background

WorldView is Shiny app, designed as a teaching tool for the PSYC382 Culture and Cognition course and is currently used for the main student research project in that course. WorldView uses data from the World Values Survey (WVS), a global research project that has collected data since 1981 on people's beliefs, values and cultural norms in over 100 countries. The most recent version of the app contains data from Wave 7 of the WVS.

Problem statement

The aim is to further develop the app so that it becomes useful in other courses and potentially for research at other universities. As such, we are looking to integrate other cross-national datasets, e.g. UN Human Development Reports data, develop visualisations and implement models to test hypotheses, relevant to healthcare, well-being and social psychology.

Example topics to explore are:

- Relationship between a country's affluence and individual well-being.
- Links between education and secularization.

Expected methodology

Students will identify, source and preprocess relevant datasets to integrate into the WorldView Shiny app. They will produce visualizations for Exploratory Data Analysis (EDA) and to communicate data features. From this, they will identify some topics and develop hypotheses for 300-level psychology students to explore, implementing models to allow the students to test their chosen hypotheses.

Deliverables

- One or more preprocessed datasets with EDA visualisations integrated into the Shiny app.
- A set of research questions/hypotheses and associated models.
- Documentation about the new datasets, covering preprocessing steps and hypotheses and models available.

Data / resources provided

GitHub repo: <https://github.com/Joseph-Watts/WorldView>

If you are interested in this project, you can contact Joseph Watts or Nicki Cartlidge (nicki.cartlidge@canterbury.ac.nz) for a link to the latest version of the WorldView app.

Skills required

- Strong R coding skills; previous experience with R Shiny is useful.
- Knowledge/experience of visualization and EDA using ggplot or plotly, including visualizing spatial data.
- Understanding of a range of modelling techniques.

References

- <https://www.worldvaluessurvey.org>
- https://en.wikipedia.org/wiki/World_Values_Survey
- <https://hdr.undp.org/data-center>

Project Code	WorldView2
Type	DATA601 (Academic research project)
Title	Develop the WorldView Shiny App to add language phylogeny and phylogenetic comparative methods
Department	School of Psychology, Speech and Hearing
Academic supervisor	Joseph Watts (joseph.watts@canterbury.ac.nz)
Location	University of Canterbury
Students	Individual

Project Summary

Background

WorldView is Shiny app, designed as a teaching tool for the PSYC382 Culture and Cognition course and is currently used for the main student research project in that course. WorldView uses data from the World Values Survey (WVS), a global research project that has collected data since 1981 on people's beliefs, values and cultural norms in over 100 countries. The most recent version of the app contains data from Wave 7 of the WVS.

Problem statement

The aim is to further develop the app to:

- Link countries to a global language phylogeny (i.e. a representation of the evolution of a language and its relationship to other languages).
- Create visualisations for variables from the WVS survey mapped onto the phylogeny.
- Add phylogenetic comparative methods, such as:
 - Estimating the extent of phylogenetic structure in the data.
 - Phylogenetic linear regression.
- Enable exploration of topics such as:
 - Stability of cultural traits over time.
 - The extent to which language predicts cultural and social values.

Expected methodology

Students will learn about phylogeny and phylogenetic comparative methods as they apply to languages. Language tree visualisations will be integrated into the WorldView Shiny app, mapping variables of interest from the WVS to the phylogeny. Topics relating to culture and social values will be defined and explored with phylogenetic methods.

Deliverables

- Language phylogeny trees integrated into the Shiny app.
- A set of research questions/topics and associated models.
- Documentation about the language phylogeny development and usage, and the topics and models available.

Data / resources provided

GitHub repo: <https://github.com/Joseph-Watts/WorldView>

If you are interested in this project, you can contact Joseph Watts or Nicki Cartlidge (nicki.cartlidge@canterbury.ac.nz) for a link to the latest version of the WorldView app.

Skills required

- Good R coding skills; previous experience with R Shiny is useful.
- Particularly suited for students with experience or interest in evolutionary biology and development of languages.

References

- <https://www.worldvaluessurvey.org>
- https://en.wikipedia.org/wiki/World_Values_Survey
- <https://hdr.undp.org/data-center>